tau+ (-15)
Particle 0
tauVisibleLeptonic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-101.331, -33.028, 304.036, 322.180)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1182 simE=1.251600
subgraph recHits: 0 missing=1182 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 34 dE=65.783913
raw GEN: GenParticle #18 key=2
raw SIM: n/a

Primary
Vertex 9
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.086, -0.027, 2.418, -0.243)
nIn: 1 nOut: 3
raw GEN: GenVertex #11 key=-11
raw SIM: SimVertex #81 key=9

tau- (15)
Particle 1
tauVisibleHadronic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (101.331, 33.028, -304.036, 322.180)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2347 simE=2.139073
subgraph recHits: 0 missing=2347 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 38 dE=91.527878
raw GEN: GenParticle #19 key=3
raw SIM: n/a

Primary
Vertex 6
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.184, 0.061, 1.609, -0.232)
nIn: 1 nOut: 3
raw GEN: GenVertex #8 key=-8
raw SIM: SimVertex #78 key=6

tau+ (-15)
Particle 2
tauVisibleLeptonic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (52.390, -68.469, -261.140, 275.009)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 55 simE=0.020110
subgraph recHits: 0 missing=55 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 42 dE=42.747112
raw GEN: GenParticle #20 key=4
raw SIM: n/a

Primary
Vertex 0
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.239, -0.314, 0.967, -0.210)
nIn: 1 nOut: 3
raw GEN: GenVertex #2 key=-2
raw SIM: SimVertex #72 key=0

tau- (15)
Particle 3
tauVisibleHadronic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-52.390, 68.469, 261.140, 275.009)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1628 simE=0.953486
subgraph recHits: 0 missing=1628 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 40 dE=88.131439
raw GEN: GenParticle #21 key=5
raw SIM: n/a

Primary
Vertex 1
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.118, 0.152, 2.745, -0.232)
nIn: 1 nOut: 3
raw GEN: GenVertex #3 key=-3
raw SIM: SimVertex #73 key=1

tau+ (-15)
Particle 4
tauVisibleLeptonic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (175.500, 127.052, -37.860, 219.952)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 318 simE=144.768707
subgraph recHits: 0 missing=318 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 18 dE=30.232281
raw GEN: GenParticle #22 key=6
raw SIM: n/a

Primary
Vertex 2
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (1.046, 0.759, 1.940, -0.208)
nIn: 1 nOut: 3
raw GEN: GenVertex #4 key=-4
raw SIM: SimVertex #74 key=2

tau- (15)
Particle 5
tauVisibleLeptonic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-175.500, -127.052, 37.860, 219.952)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 8 simE=0.310979
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 16 dE=29.742107
raw GEN: GenParticle #23 key=7
raw SIM: n/a

Primary
Vertex 3
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-1.300, -0.939, 2.446, -0.198)
nIn: 1 nOut: 3
raw GEN: GenVertex #5 key=-5
raw SIM: SimVertex #75 key=3

tau+ (-15)
Particle 6
tauVisibleLeptonic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-104.424, -259.847, -237.507, 367.202)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 281 simE=88.391411
subgraph recHits: 0 missing=281 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 20 dE=37.399170
raw GEN: GenParticle #24 key=8
raw SIM: n/a

Primary
Vertex 4
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.557, -1.382, 0.904, -0.187)
nIn: 1 nOut: 3
raw GEN: GenVertex #6 key=-6
raw SIM: SimVertex #76 key=4

tau- (15)
Particle 7
tauVisibleHadronic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (104.424, 259.847, 237.507, 367.202)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 704 simE=120.435997
subgraph recHits: 0 missing=704 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 45 dE=175.563843
raw GEN: GenParticle #25 key=9
raw SIM: n/a

Primary
Vertex 5
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.274, 0.688, 2.795, -0.220)
nIn: 1 nOut: 3
raw GEN: GenVertex #7 key=-7
raw SIM: SimVertex #77 key=5

tau+ (-15)
Particle 8
tauVisibleHadronic
signal
pid: tau+ (-15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (33.832, 72.137, -417.384, 424.924)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2258 simE=1.065101
subgraph recHits: 0 missing=2258 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 33 dE=64.941605
raw GEN: GenParticle #26 key=10
raw SIM: n/a

Primary
Vertex 7
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.144, 0.313, 0.359, -0.191)
nIn: 1 nOut: 3
raw GEN: GenVertex #9 key=-9
raw SIM: SimVertex #79 key=7

tau- (15)
Particle 9
tauVisibleHadronic
signal
pid: tau- (15)
status: 2
statusFlags: 12291
flags: isPrompt, isDecayedLeptonHadron, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-33.832, -72.137, 417.384, 424.924)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3552 simE=1.710313
subgraph recHits: 0 missing=3552 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 46 dE=163.183121
raw GEN: GenParticle #27 key=11
raw SIM: n/a

Primary
Vertex 8
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.016, -0.030, 2.343, -0.246)
nIn: 1 nOut: 4
raw GEN: GenVertex #10 key=-10
raw SIM: SimVertex #80 key=8

anti-nu_tau (-16)
Particle 10
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (29.676, -39.940, -151.538, 159.499)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #29 key=12
raw SIM: n/a

mu+ (-13)
Particle 11
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: mu+ (-13)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (17.725, -22.581, -86.700, 91.328)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 55 simE=0.020110
direct recHits: 0 missing=55 recoE=0.000000
subgraph simHits: 55 simE=0.020110
subgraph recHits: 0 missing=55 recoE=0.000000
direct tracker simHits: 42 dE=42.747112
subgraph tracker simHits: 42 dE=42.747112
checkpointId: 0
x4@checkpoint: (58.942, -78.167, -293.500, 0.000)
p4@checkpoint: (16.845, -23.237, -86.666, 91.295)
raw GEN: GenParticle #30 key=13
raw SIM: SimTrack #371 key=2

nu_mu (14)
Particle 12
stableLegsFromInitialState
stableDecayProducts
pid: nu_mu (14)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (4.988, -5.949, -22.902, 24.182)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #31 key=14
raw SIM: n/a

nu_tau (16)
Particle 13
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-12.430, 15.577, 62.087, 65.207)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #32 key=15
raw SIM: n/a

pi0 (111)
Particle 14
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-2.416, 3.350, 12.486, 13.152)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 296 simE=0.101704
subgraph recHits: 0 missing=296 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 31 dE=59.455719
raw GEN: GenParticle #33 key=16
raw SIM: n/a

Primary
Vertex 10
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.118, 0.152, 2.745, -0.232)
nIn: 1 nOut: 2
raw GEN: GenVertex #12 key=-12
raw SIM: SimVertex #82 key=10

pi- (-211)
Particle 15
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-37.544, 49.542, 186.568, 196.650)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 1327 simE=0.833277
direct recHits: 0 missing=1327 recoE=0.000000
subgraph simHits: 1333 simE=0.851782
subgraph recHits: 0 missing=1333 recoE=0.000000
direct tracker simHits: 28 dE=28.675720
subgraph tracker simHits: 20 dE=28.675720
checkpointId: 0
x4@checkpoint: (-59.301, 76.859, 293.500, 0.000)
p4@checkpoint: (-38.391, 48.893, 186.542, 196.627)
raw GEN: GenParticle #34 key=17
raw SIM: SimTrack #372 key=5

HadronInelastic

Vertex 210
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-70.393, -91.956, 140.419, 46.013)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #368 key=296

HadronInelastic

Vertex 211
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-70.139, -95.643, 136.486, 46.669)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #369 key=297

anti-nu_tau (-16)
Particle 16
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (34.018, 25.052, -6.795, 42.790)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #35 key=18
raw SIM: n/a

e+ (-11)
Particle 17
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e+ (-11)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (117.437, 84.708, -25.619, 147.048)
nProdVtx: 1 nDecayVtx: 6
nParents: 1 nChildren: 6
nCheckpoints: 1
direct simHits: 226 simE=59.951153
direct recHits: 0 missing=226 recoE=0.000000
subgraph simHits: 318 simE=144.768707
subgraph recHits: 0 missing=318 recoE=0.000000
direct tracker simHits: 30 dE=30.232279
subgraph tracker simHits: 18 dE=30.232281
checkpointId: 0
x4@checkpoint: (100.665, 71.200, -19.660, 0.000)
p4@checkpoint: (49.571, 34.223, -10.663, 61.174)
raw GEN: GenParticle #36 key=19
raw SIM: SimTrack #373 key=7

Bremsstrahlung
Vertex 194
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (11.736, 8.461, -0.391, 0.238)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #351 key=279

Bremsstrahlung
Vertex 195
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.952, 12.214, -1.528, 0.456)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #352 key=280

Bremsstrahlung
Vertex 196
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.320, 66.813, -18.295, 3.663)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #353 key=281

Bremsstrahlung
Vertex 197
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.479, 66.923, -18.329, 3.670)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #354 key=282

Bremsstrahlung
Vertex 198
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.528, 66.957, -18.340, 3.672)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #355 key=283

Bremsstrahlung
Vertex 199
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.547, 66.970, -18.344, 3.673)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #356 key=284

nu_e (12)
Particle 18
stableLegsFromInitialState
stableDecayProducts
pid: nu_e (12)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (24.045, 17.291, -5.447, 30.114)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #37 key=20
raw SIM: n/a

nu_tau (16)
Particle 19
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-72.781, -52.649, 16.502, 91.330)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #38 key=21
raw SIM: n/a

mu- (13)
Particle 20
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: mu- (13)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-43.319, -31.705, 8.984, 54.429)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 8 simE=0.310979
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.310979
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 29 dE=29.742105
subgraph tracker simHits: 16 dE=29.742107
checkpointId: 0
x4@checkpoint: (-98.570, -74.073, 22.813, 0.000)
p4@checkpoint: (-42.468, -32.792, 8.984, 54.402)
raw GEN: GenParticle #39 key=22
raw SIM: SimTrack #374 key=10

Ionisation

Vertex 191
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-346.735, -277.108, 77.537, 14.769)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #348 key=276

Ionisation

Vertex 192
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-478.581, -386.824, 106.838, 20.573)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #349 key=277

Ionisation

Vertex 193
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-498.011, -402.845, 111.149, 21.426)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #350 key=278

anti-nu_mu (-14)
Particle 21
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_mu (-14)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-59.400, -42.698, 12.374, 74.193)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #40 key=23
raw SIM: n/a

anti-nu_tau (-16)
Particle 22
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-20.375, -51.880, -46.871, 72.826)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #41 key=24
raw SIM: n/a

e+ (-11)
Particle 23
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e+ (-11)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-25.869, -62.923, -58.153, 89.500)
nProdVtx: 1 nDecayVtx: 20
nParents: 1 nChildren: 21
nCheckpoints: 1
direct simHits: 192 simE=52.445778
direct recHits: 0 missing=192 recoE=0.000000
subgraph simHits: 281 simE=88.391411
subgraph recHits: 0 missing=281 recoE=0.000000
direct tracker simHits: 37 dE=37.399174
subgraph tracker simHits: 20 dE=37.399170
checkpointId: 0
x4@checkpoint: (-48.029, -113.561, -103.234, 0.000)
p4@checkpoint: (-16.122, -36.974, -34.491, 53.072)
raw GEN: GenParticle #42 key=25
raw SIM: SimTrack #375 key=13

Bremsstrahlung
Vertex 170
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-15.248, -36.792, -31.867, 1.495)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #325 key=253

Bremsstrahlung
Vertex 171
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.503, -51.676, -45.668, 2.204)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #326 key=254

Bremsstrahlung
Vertex 172
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-21.979, -52.805, -46.716, 2.258)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #327 key=255

Bremsstrahlung
Vertex 173
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-34.227, -81.606, -73.472, 3.631)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #328 key=256

Bremsstrahlung
Vertex 174
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-34.247, -81.653, -73.516, 3.633)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #329 key=257

Bremsstrahlung
Vertex 175
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-44.821, -106.184, -96.356, 4.806)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #330 key=258

Bremsstrahlung
Vertex 176
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-44.836, -106.219, -96.388, 4.807)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #331 key=259

Bremsstrahlung
Vertex 177
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-44.943, -106.465, -96.618, 4.819)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #332 key=260

Bremsstrahlung
Vertex 178
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-45.001, -106.599, -96.743, 4.825)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #333 key=261

Bremsstrahlung
Vertex 179
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-45.013, -106.626, -96.768, 4.827)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #334 key=262

Bremsstrahlung
Vertex 180
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-46.423, -109.873, -99.794, 4.982)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #335 key=263

Annihilation

Vertex 181
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-17.844, -116.913, -119.577, 6.954)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #336 key=264

Bremsstrahlung
Vertex 183
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-44.365, -106.886, -96.744, 4.826)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #340 key=268

Bremsstrahlung
Vertex 184
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-44.367, -106.890, -96.748, 4.826)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #341 key=269

Bremsstrahlung
Vertex 185
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-44.643, -107.436, -97.330, 4.854)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #342 key=270

Bremsstrahlung
Vertex 186
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-45.308, -106.802, -101.112, 5.276)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #343 key=271

Bremsstrahlung
Vertex 187
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-45.703, -107.591, -101.220, 5.307)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #344 key=272

Bremsstrahlung
Vertex 188
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-46.700, -106.500, -102.798, 5.409)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #345 key=273

Bremsstrahlung
Vertex 189
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-46.002, -106.244, -103.002, 5.437)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #346 key=274

Annihilation

Vertex 190
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-46.336, -106.029, -102.496, 5.462)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #347 key=275

nu_e (12)
Particle 24
stableLegsFromInitialState
stableDecayProducts
pid: nu_e (12)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-58.180, -145.044, -132.483, 204.877)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #43 key=26
raw SIM: n/a

nu_tau (16)
Particle 25
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (63.490, 156.498, 143.044, 221.323)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #44 key=27
raw SIM: n/a

pi0 (111)
Particle 26
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (34.003, 85.940, 78.907, 121.525)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 613 simE=107.572792
subgraph recHits: 0 missing=613 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 38 dE=114.135307
raw GEN: GenParticle #45 key=28
raw SIM: n/a

Primary
Vertex 11
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.275, 0.689, 2.796, -0.220)
nIn: 1 nOut: 2
raw GEN: GenVertex #13 key=-13
raw SIM: SimVertex #83 key=11

pi- (-211)
Particle 27
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (6.931, 17.409, 15.557, 24.354)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 12
nCheckpoints: 1
direct simHits: 130 simE=12.657404
direct recHits: 0 missing=130 recoE=0.000000
subgraph simHits: 186 simE=12.863174
subgraph recHits: 0 missing=186 recoE=0.000000
direct tracker simHits: 61 dE=61.428520
subgraph tracker simHits: 19 dE=61.428520
checkpointId: 0
x4@checkpoint: (41.326, 116.168, 104.525, 0.000)
p4@checkpoint: (5.555, 17.733, 15.438, 24.159)
raw GEN: GenParticle #46 key=29
raw SIM: SimTrack #376 key=16

Ionisation

Vertex 164
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (40.099, 112.269, 101.129, 4.917)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #319 key=247

HadronInelastic

Vertex 165
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.284, 67.334, 131.736, 127.571)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #320 key=248

HadronInelastic

Vertex 166
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (89.161, 78.337, 184.218, 11.001)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #321 key=249

HadronInelastic

Vertex 167
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (88.377, 79.002, 185.180, 11.372)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #322 key=250

HadronInelastic

Vertex 168
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (110.563, 40.330, 197.595, 19.042)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #323 key=251

nu_tau (16)
Particle 28
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (5.085, 1.678, -16.341, 17.196)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #47 key=30
raw SIM: n/a

pi0 (111)
Particle 29
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (69.799, 22.997, -208.203, 220.792)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1565 simE=1.548358
subgraph recHits: 0 missing=1565 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 30 dE=56.132904
raw GEN: GenParticle #48 key=31
raw SIM: n/a

Primary
Vertex 12
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.185, 0.061, 1.604, -0.232)
nIn: 1 nOut: 2
raw GEN: GenVertex #14 key=-14
raw SIM: SimVertex #84 key=12

pi- (-211)
Particle 30
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (26.447, 8.353, -79.492, 84.192)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 1128 simE=0.589661
direct recHits: 0 missing=1128 recoE=0.000000
subgraph simHits: 1131 simE=0.590718
subgraph recHits: 0 missing=1131 recoE=0.000000
direct tracker simHits: 35 dE=35.394981
subgraph tracker simHits: 22 dE=35.394981
checkpointId: 0
x4@checkpoint: (97.711, 33.114, -293.500, 0.000)
p4@checkpoint: (26.076, 9.437, -79.457, 84.157)
raw GEN: GenParticle #49 key=32
raw SIM: SimTrack #377 key=18

HadronInelastic

Vertex 163
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-1.609, -116.694, -222.189, 144.286)
nIn: 1 nOut: 3
raw GEN: n/a
raw SIM: SimVertex #318 key=246

anti-nu_tau (-16)
Particle 31
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (17.574, 35.538, -207.991, 211.735)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #50 key=33
raw SIM: n/a

pi0 (111)
Particle 32
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (4.619, 10.315, -58.248, 59.335)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 747 simE=0.349267
subgraph recHits: 0 missing=747 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #51 key=34
raw SIM: n/a

Primary
Vertex 13
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (0.144, 0.313, 0.358, -0.191)
nIn: 1 nOut: 2
raw GEN: GenVertex #15 key=-15
raw SIM: SimVertex #85 key=13

pi+ (211)
Particle 33
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (11.640, 26.284, -151.145, 153.854)
nProdVtx: 1 nDecayVtx: 6
nParents: 1 nChildren: 12
nCheckpoints: 1
direct simHits: 1750 simE=0.660544
direct recHits: 0 missing=1750 recoE=0.000000
subgraph simHits: 1786 simE=0.715835
subgraph recHits: 0 missing=1786 recoE=0.000000
direct tracker simHits: 59 dE=59.941605
subgraph tracker simHits: 33 dE=64.941605
checkpointId: 0
x4@checkpoint: (23.314, 51.165, -293.500, 0.000)
p4@checkpoint: (12.185, 26.016, -151.042, 153.750)
raw GEN: GenParticle #52 key=35
raw SIM: SimTrack #378 key=20

Ionisation

Vertex 157
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.479, 12.273, -68.506, 2.148)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #305 key=233

HadronInelastic

Vertex 158
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (113.413, -22.959, -230.971, 21.879)
nIn: 1 nOut: 5
raw GEN: n/a
raw SIM: SimVertex #306 key=234

HadronInelastic

Vertex 159
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-73.512, -80.754, -219.254, 43.902)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #308 key=236

HadronInelastic

Vertex 160
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (11.787, 12.182, -43.644, 25.793)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #311 key=239

Bremsstrahlung
Vertex 161
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (18.132, 36.762, -262.641, 8.804)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #312 key=240

Bremsstrahlung
Vertex 162
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (1.982, 4.737, -24.897, 0.671)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #317 key=245

nu_tau (16)
Particle 34
stableLegsFromInitialState
stableDecayProducts
pid: nu_tau (16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.760, -1.396, 7.776, 7.937)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #53 key=36
raw SIM: n/a

pi0 (111)
Particle 35
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-13.246, -27.549, 158.218, 161.144)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1486 simE=0.973936
subgraph recHits: 0 missing=1486 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 20 dE=46.961910
raw GEN: GenParticle #54 key=37
raw SIM: n/a

Primary
Vertex 14
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.016, -0.030, 2.344, -0.246)
nIn: 1 nOut: 2
raw GEN: GenVertex #16 key=-16
raw SIM: SimVertex #86 key=14

pi0 (111)
Particle 36
reconstructableFromSignal
reconstructableFinalState
pid: pi0 (111)
status: 2
statusFlags: 28734
flags: isDecayedLeptonHadron, isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-3.811, -7.790, 46.617, 47.417)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 922 simE=0.280971
subgraph recHits: 0 missing=922 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 25 dE=71.432343
raw GEN: GenParticle #55 key=38
raw SIM: n/a

Primary
Vertex 15
domain: GEN+SIM
reason: Primary
genEvent: 0
hasGen: yes hasSim: yes
isSource: no isSink: no
x4: (-0.017, -0.031, 2.344, -0.246)
nIn: 1 nOut: 2
raw GEN: GenVertex #17 key=-17
raw SIM: SimVertex #87 key=15

pi- (-211)
Particle 37
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-16.016, -35.402, 204.773, 208.427)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1334 simE=0.455407
direct recHits: 0 missing=1334 recoE=0.000000
subgraph simHits: 1334 simE=0.455407
subgraph recHits: 0 missing=1334 recoE=0.000000
direct tracker simHits: 44 dE=44.788891
subgraph tracker simHits: 44 dE=44.788891
checkpointId: 0
x4@checkpoint: (-22.387, -50.535, 293.500, 0.000)
p4@checkpoint: (-15.454, -35.630, 204.702, 208.353)
raw GEN: GenParticle #56 key=39
raw SIM: SimTrack #379 key=22

anti-nu_tau (-16)
Particle 38
stableLegsFromInitialState
stableDecayProducts
pid: anti-nu_tau (-16)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-27.106, -8.627, 80.744, 85.608)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #57 key=40
raw SIM: n/a

e+ (-11)
Particle 39
stableLegsFromInitialState
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e+ (-11)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-47.915, -16.457, 145.300, 153.879)
nProdVtx: 1 nDecayVtx: 7
nParents: 1 nChildren: 7
nCheckpoints: 1
direct simHits: 602 simE=0.399199
direct recHits: 0 missing=602 recoE=0.000000
subgraph simHits: 1182 simE=1.251600
subgraph recHits: 0 missing=1182 recoE=0.000000
direct tracker simHits: 46 dE=47.331375
subgraph tracker simHits: 34 dE=65.783913
checkpointId: 0
x4@checkpoint: (-96.711, -30.926, 293.500, 0.000)
p4@checkpoint: (-18.497, -5.358, 55.267, 58.525)
raw GEN: GenParticle #58 key=41
raw SIM: SimTrack #380 key=24

Bremsstrahlung
Vertex 146
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-19.777, -6.734, 62.086, 1.865)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #284 key=212

Bremsstrahlung
Vertex 147
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-23.140, -7.870, 72.266, 2.224)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #285 key=213

Bremsstrahlung
Vertex 148
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-23.551, -8.008, 73.510, 2.268)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #286 key=214

Bremsstrahlung
Vertex 149
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-23.569, -8.014, 73.563, 2.270)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #287 key=215

Bremsstrahlung
Vertex 150
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-23.582, -8.019, 73.601, 2.271)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #288 key=216

Bremsstrahlung
Vertex 151
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-38.252, -12.898, 117.940, 3.838)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #289 key=217

Bremsstrahlung
Vertex 152
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-42.845, -14.397, 131.799, 4.327)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #290 key=218

nu_e (12)
Particle 40
stableLegsFromInitialState
stableDecayProducts
pid: nu_e (12)
status: 1
statusFlags: 12348
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectTauDecayProduct, isDirectPromptTauDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-26.309, -7.944, 77.992, 82.693)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #59 key=42
raw SIM: n/a

gamma (22)
Particle 41
stableLegsFromInitialState
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-0.749, 1.140, 4.190, 4.406)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 100 simE=0.036131
subgraph recHits: 0 missing=100 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-52.074, 79.290, 293.500, 0.000)
p4@checkpoint: (-0.749, 1.140, 4.190, 4.406)
raw GEN: GenParticle #60 key=43
raw SIM: SimTrack #381 key=26

PairConversion

Vertex 145
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-52.870, 80.502, 297.953, 10.124)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #282 key=210

gamma (22)
Particle 42
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-1.667, 2.210, 8.296, 8.746)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 196 simE=0.065573
subgraph recHits: 0 missing=196 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 31 dE=59.455719
raw GEN: GenParticle #61 key=44
raw SIM: SimTrack #382 key=27

PairConversion

Vertex 139
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-14.423, 19.109, 73.924, 2.271)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #268 key=196

gamma (22)
Particle 43
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (30.342, 76.701, 70.368, 108.422)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 557 simE=98.238075
subgraph recHits: 0 missing=557 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 38 dE=114.135307
raw GEN: GenParticle #62 key=45
raw SIM: SimTrack #383 key=28

PairConversion

Vertex 65
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.687, 6.787, 8.390, 0.068)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #179 key=107

gamma (22)
Particle 44
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (3.661, 9.239, 8.539, 13.102)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 198 simE=9.334736
subgraph recHits: 0 missing=198 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #63 key=46
raw SIM: SimTrack #384 key=29

PairConversion

Vertex 51
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.637, 107.599, 101.609, 4.838)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #165 key=93

gamma (22)
Particle 45
stableLegsFromInitialState
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (56.820, 18.749, -169.648, 179.890)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1018 simE=1.194705
subgraph recHits: 0 missing=1018 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (99.024, 32.675, -293.500, 0.000)
p4@checkpoint: (56.820, 18.749, -169.648, 179.890)
raw GEN: GenParticle #64 key=47
raw SIM: SimTrack #385 key=30

PairConversion

Vertex 50
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (109.684, 36.192, -325.328, 11.331)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #164 key=92

gamma (22)
Particle 46
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (12.979, 4.248, -38.555, 40.902)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 665 simE=0.353654
subgraph recHits: 0 missing=665 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 30 dE=56.132904
raw GEN: GenParticle #65 key=48
raw SIM: SimTrack #386 key=31

PairConversion

Vertex 37
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (15.824, 5.180, -44.852, 1.412)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #140 key=68

gamma (22)
Particle 47
stableLegsFromInitialState
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (1.035, 2.450, -13.759, 14.013)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 252 simE=0.080721
subgraph recHits: 0 missing=252 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (22.259, 52.647, -293.500, 0.000)
p4@checkpoint: (1.035, 2.450, -13.759, 14.013)
raw GEN: GenParticle #66 key=49
raw SIM: SimTrack #387 key=32

PairConversion

Vertex 36
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (22.567, 53.376, -297.593, 9.932)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #139 key=67

gamma (22)
Particle 48
stableLegsFromInitialState
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (3.584, 7.865, -44.490, 45.321)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 588 simE=0.268545
subgraph recHits: 0 missing=588 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (23.816, 52.261, -293.500, 0.000)
p4@checkpoint: (3.584, 7.865, -44.490, 45.321)
raw GEN: GenParticle #67 key=50
raw SIM: SimTrack #388 key=33

PairConversion

Vertex 35
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (24.125, 52.941, -297.347, 9.925)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #133 key=61

gamma (22)
Particle 49
stableLegsFromInitialState
stableDecayProducts
caloBoundary
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-9.586, -20.052, 115.212, 117.337)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1043 simE=0.703788
subgraph recHits: 0 missing=1043 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-24.241, -50.704, 293.500, 0.000)
p4@checkpoint: (-9.586, -20.052, 115.212, 117.337)
raw GEN: GenParticle #68 key=51
raw SIM: SimTrack #389 key=34

PairConversion

Vertex 32
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-26.546, -55.525, 321.198, 10.586)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #130 key=58

gamma (22)
Particle 50
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-3.660, -7.497, 43.006, 43.808)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 671 simE=0.270147
subgraph recHits: 0 missing=671 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 20 dE=46.961910
raw GEN: GenParticle #69 key=52
raw SIM: SimTrack #390 key=35

PairConversion

Vertex 28
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-10.801, -22.120, 129.062, 4.060)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #116 key=44

gamma (22)
Particle 51
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-2.086, -4.369, 25.754, 26.205)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 547 simE=0.162058
subgraph recHits: 0 missing=547 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 23 dE=41.656631
raw GEN: GenParticle #70 key=53
raw SIM: SimTrack #391 key=36

PairConversion

Vertex 21
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-7.725, -16.174, 97.512, 2.984)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #98 key=26

gamma (22)
Particle 52
stableLegsFromInitialState
stableDecayProducts
pid: gamma (22)
status: 1
statusFlags: 12364
flags: isTauDecayProduct, isPromptTauDecayProduct, isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-1.725, -3.421, 20.863, 21.212)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 390 simE=0.118912
subgraph recHits: 0 missing=390 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 15 dE=29.775707
raw GEN: GenParticle #71 key=54
raw SIM: SimTrack #392 key=37

PairConversion

Vertex 16
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.016, -15.898, 99.099, 3.035)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #88 key=16

e- (11)
Particle 53
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.297, -2.573, 15.690, 15.952)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 242 simE=0.081461
direct recHits: 0 missing=242 recoE=0.000000
subgraph simHits: 266 simE=0.085211
subgraph recHits: 0 missing=266 recoE=0.000000
direct tracker simHits: 14 dE=14.765423
subgraph tracker simHits: 12 dE=14.765423
checkpointId: 0
x4@checkpoint: (-21.665, -48.817, 293.500, 0.000)
p4@checkpoint: (-0.832, -2.501, 14.364, 14.604)
raw GEN: n/a
raw SIM: SimTrack #393 key=38

Bremsstrahlung
Vertex 18
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-8.099, -16.063, 100.104, 3.069)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #94 key=22

Bremsstrahlung
Vertex 19
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-9.376, -18.647, 115.820, 3.602)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #95 key=23

Bremsstrahlung
Vertex 20
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-12.824, -26.042, 160.312, 5.111)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #96 key=24

e+ (-11)
Particle 54
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.429, -0.849, 5.173, 5.259)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 112 simE=0.032183
direct recHits: 0 missing=112 recoE=0.000000
subgraph simHits: 125 simE=0.033701
subgraph recHits: 0 missing=125 recoE=0.000000
direct tracker simHits: 15 dE=15.010284
subgraph tracker simHits: 11 dE=15.010284
checkpointId: 0
x4@checkpoint: (-30.703, -43.163, 293.500, 0.000)
p4@checkpoint: (-0.681, -0.515, 4.667, 4.744)
raw GEN: n/a
raw SIM: SimTrack #394 key=39

Bremsstrahlung
Vertex 17
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-11.148, -21.541, 134.142, 4.224)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #91 key=19

gamma (22)
Particle 55
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.045, -0.074, 0.470, 0.478)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.001518
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.001518
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-26.416, -46.705, 293.500, 0.000)
p4@checkpoint: (-0.045, -0.074, 0.470, 0.478)
raw GEN: n/a
raw SIM: SimTrack #397 key=44

gamma (22)
Particle 56
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.063, -0.126, 0.767, 0.780)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.002881
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.002881
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-24.026, -47.795, 293.500, 0.000)
p4@checkpoint: (-0.063, -0.126, 0.767, 0.780)
raw GEN: n/a
raw SIM: SimTrack #400 key=8697

gamma (22)
Particle 57
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.024, -0.050, 0.305, 0.310)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000317
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000317
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-23.635, -47.944, 293.500, 0.000)
p4@checkpoint: (-0.024, -0.050, 0.305, 0.310)
raw GEN: n/a
raw SIM: SimTrack #401 key=8698

gamma (22)
Particle 58
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, -0.037, 0.223, 0.227)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000552
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.000552
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-22.816, -48.368, 293.500, 0.000)
p4@checkpoint: (-0.017, -0.037, 0.223, 0.227)
raw GEN: n/a
raw SIM: SimTrack #402 key=8700

e- (11)
Particle 59
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.684, -1.432, 8.442, 8.590)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 78 simE=0.018664
direct recHits: 0 missing=78 recoE=0.000000
subgraph simHits: 222 simE=0.051962
subgraph recHits: 0 missing=222 recoE=0.000000
direct tracker simHits: 15 dE=15.721412
subgraph tracker simHits: 19 dE=24.955519
checkpointId: 0
x4@checkpoint: (-13.982, -51.599, 293.500, 0.000)
p4@checkpoint: (0.101, -0.501, 2.765, 2.812)
raw GEN: n/a
raw SIM: SimTrack #404 key=35419

Bremsstrahlung
Vertex 23
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-9.011, -18.931, 113.746, 3.535)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #107 key=35

Bremsstrahlung
Vertex 24
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-11.842, -25.677, 152.640, 4.855)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #108 key=36

Bremsstrahlung
Vertex 25
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-12.553, -27.650, 163.775, 5.233)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #110 key=38

Bremsstrahlung
Vertex 26
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-13.686, -31.683, 185.950, 5.986)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #111 key=39

e+ (-11)
Particle 60
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.403, -2.936, 17.312, 17.615)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 293 simE=0.099174
direct recHits: 0 missing=293 recoE=0.000000
subgraph simHits: 325 simE=0.110097
subgraph recHits: 0 missing=325 recoE=0.000000
direct tracker simHits: 16 dE=16.701109
subgraph tracker simHits: 14 dE=16.701107
checkpointId: 0
x4@checkpoint: (-25.643, -48.121, 293.500, 0.000)
p4@checkpoint: (-1.581, -2.422, 15.488, 15.756)
raw GEN: n/a
raw SIM: SimTrack #405 key=35420

Bremsstrahlung
Vertex 22
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-12.365, -25.452, 152.883, 4.863)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #101 key=29

gamma (22)
Particle 61
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.154, -0.295, 1.779, 1.810)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 47 simE=0.010922
direct recHits: 0 missing=47 recoE=0.000000
subgraph simHits: 47 simE=0.010922
subgraph recHits: 0 missing=47 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-24.551, -48.788, 293.500, 0.000)
p4@checkpoint: (-0.154, -0.295, 1.779, 1.810)
raw GEN: n/a
raw SIM: SimTrack #408 key=35425

gamma (22)
Particle 62
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.007, 0.041, 0.042)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000242
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000242
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-22.881, -49.621, 293.500, 0.000)
p4@checkpoint: (-0.003, -0.007, 0.041, 0.042)
raw GEN: n/a
raw SIM: SimTrack #414 key=64915

gamma (22)
Particle 63
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.363, -0.933, 5.322, 5.415)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 132 simE=0.029948
subgraph recHits: 0 missing=132 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 6 dE=9.234106
raw GEN: n/a
raw SIM: SimTrack #415 key=64918

PairConversion

Vertex 27
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-16.750, -38.281, 224.512, 7.294)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #115 key=43

gamma (22)
Particle 64
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, -0.043, 0.242, 0.246)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.001868
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.001868
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-20.344, -50.804, 293.500, 0.000)
p4@checkpoint: (-0.015, -0.043, 0.242, 0.246)
raw GEN: n/a
raw SIM: SimTrack #417 key=64921

gamma (22)
Particle 65
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, -0.007, 0.039, 0.040)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.001239
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.001239
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-18.318, -51.450, 293.500, 0.000)
p4@checkpoint: (-0.002, -0.007, 0.039, 0.040)
raw GEN: n/a
raw SIM: SimTrack #418 key=64922

e- (11)
Particle 66
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.347, -0.893, 5.091, 5.180)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 124 simE=0.028456
direct recHits: 0 missing=124 recoE=0.000000
subgraph simHits: 124 simE=0.028456
subgraph recHits: 0 missing=124 recoE=0.000000
direct tracker simHits: 4 dE=4.205400
subgraph tracker simHits: 4 dE=4.205400
checkpointId: 0
x4@checkpoint: (-20.528, -50.667, 293.500, 0.000)
p4@checkpoint: (-0.209, -0.931, 5.088, 5.177)
raw GEN: n/a
raw SIM: SimTrack #422 key=69841

e+ (-11)
Particle 67
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.017, -0.041, 0.230, 0.234)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.001492
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.001492
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 5 dE=5.028707
subgraph tracker simHits: 5 dE=5.028707
checkpointId: 0
x4@checkpoint: (-23.909, -34.904, 293.500, 0.000)
p4@checkpoint: (0.022, 0.037, 0.229, 0.232)
raw GEN: n/a
raw SIM: SimTrack #423 key=69842

e- (11)
Particle 68
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-2.592, -5.310, 30.455, 31.023)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 414 simE=0.199768
direct recHits: 0 missing=414 recoE=0.000000
subgraph simHits: 414 simE=0.199768
subgraph recHits: 0 missing=414 recoE=0.000000
direct tracker simHits: 10 dE=10.364159
subgraph tracker simHits: 10 dE=10.364159
checkpointId: 0
x4@checkpoint: (-23.940, -51.187, 293.500, 0.000)
p4@checkpoint: (-2.279, -5.450, 30.445, 31.013)
raw GEN: n/a
raw SIM: SimTrack #424 key=78893

e+ (-11)
Particle 69
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.068, -2.188, 12.551, 12.785)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 231 simE=0.067721
direct recHits: 0 missing=231 recoE=0.000000
subgraph simHits: 259 simE=0.070379
subgraph recHits: 0 missing=259 recoE=0.000000
direct tracker simHits: 10 dE=9.619332
subgraph tracker simHits: 20 dE=36.597755
checkpointId: 0
x4@checkpoint: (-26.874, -49.574, 293.500, 0.000)
p4@checkpoint: (-1.258, -1.827, 11.460, 11.672)
raw GEN: n/a
raw SIM: SimTrack #425 key=78894

Bremsstrahlung
Vertex 29
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-13.133, -26.754, 155.818, 4.969)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #117 key=45

Bremsstrahlung
Vertex 30
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-19.401, -37.969, 222.241, 7.225)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #120 key=48

gamma (22)
Particle 70
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.092, -0.178, 1.034, 1.053)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 27 simE=0.002443
subgraph recHits: 0 missing=27 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 16 dE=26.978422
raw GEN: n/a
raw SIM: SimTrack #426 key=78895

PairConversion

Vertex 31
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-13.135, -26.759, 155.848, 4.970)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #126 key=54

gamma (22)
Particle 71
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, -0.006, 0.036, 0.036)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000214
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000214
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-26.513, -49.816, 293.500, 0.000)
p4@checkpoint: (-0.004, -0.006, 0.036, 0.036)
raw GEN: n/a
raw SIM: SimTrack #429 key=78898

e- (11)
Particle 72
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.061, -0.116, 0.678, 0.690)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 25 simE=0.001888
direct recHits: 0 missing=25 recoE=0.000000
subgraph simHits: 25 simE=0.001888
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 22 dE=20.639837
subgraph tracker simHits: 22 dE=20.639837
checkpointId: 0
x4@checkpoint: (-0.524, -43.591, 293.500, 0.000)
p4@checkpoint: (0.120, 0.030, 0.658, 0.670)
raw GEN: n/a
raw SIM: SimTrack #435 key=98463

e+ (-11)
Particle 73
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.031, -0.062, 0.356, 0.363)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000555
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000555
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 8 dE=6.338584
subgraph tracker simHits: 8 dE=6.338584
checkpointId: 0
x4@checkpoint: (-19.612, -19.354, 293.500, 0.000)
p4@checkpoint: (0.058, -0.001, 0.349, 0.354)
raw GEN: n/a
raw SIM: SimTrack #436 key=98464

e- (11)
Particle 74
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-5.542, -11.594, 66.614, 67.842)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 706 simE=0.394761
direct recHits: 0 missing=706 recoE=0.000000
subgraph simHits: 709 simE=0.395494
subgraph recHits: 0 missing=709 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-26.546, -55.526, 321.201, 0.000)
p4@checkpoint: (-5.542, -11.592, 66.601, 67.830)
raw GEN: n/a
raw SIM: SimTrack #440 key=150510

Annihilation

Vertex 33
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.724, -69.159, 236.314, 15.010)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #131 key=59

Bremsstrahlung
Vertex 34
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-93.533, -69.214, 236.360, 15.001)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #132 key=60

e+ (-11)
Particle 75
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-4.043, -8.458, 48.598, 49.494)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 590 simE=0.308294
direct recHits: 0 missing=590 recoE=0.000000
subgraph simHits: 590 simE=0.308294
subgraph recHits: 0 missing=590 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-26.547, -55.528, 321.215, 0.000)
p4@checkpoint: (-0.055, -0.115, 0.660, 0.673)
raw GEN: n/a
raw SIM: SimTrack #441 key=150511

gamma (22)
Particle 76
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000312
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000312
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (107.506, 60.377, 210.106, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #442 key=321346

gamma (22)
Particle 77
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000339
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000339
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-100.759, -71.067, 238.845, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #443 key=321347

gamma (22)
Particle 78
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000082
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000082
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-102.012, -69.256, 232.470, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #444 key=321348

e- (11)
Particle 79
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.177, 2.581, -14.606, 14.879)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 254 simE=0.091780
direct recHits: 0 missing=254 recoE=0.000000
subgraph simHits: 254 simE=0.091780
subgraph recHits: 0 missing=254 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (24.146, 52.986, -297.602, 0.000)
p4@checkpoint: (1.168, 2.582, -14.606, 14.878)
raw GEN: n/a
raw SIM: SimTrack #445 key=349111

e+ (-11)
Particle 80
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (2.407, 5.284, -29.884, 30.443)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 452 simE=0.176765
direct recHits: 0 missing=452 recoE=0.000000
subgraph simHits: 452 simE=0.176765
subgraph recHits: 0 missing=452 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (24.154, 53.005, -297.706, 0.000)
p4@checkpoint: (2.260, 4.967, -28.084, 28.609)
raw GEN: n/a
raw SIM: SimTrack #446 key=349112

e- (11)
Particle 81
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.261, 0.616, -3.466, 3.530)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 99 simE=0.022903
direct recHits: 0 missing=99 recoE=0.000000
subgraph simHits: 99 simE=0.022903
subgraph recHits: 0 missing=99 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (22.606, 53.469, -298.116, 0.000)
p4@checkpoint: (0.267, 0.621, -3.460, 3.526)
raw GEN: n/a
raw SIM: SimTrack #453 key=424968

e+ (-11)
Particle 82
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.774, 1.834, -10.293, 10.484)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 191 simE=0.057819
direct recHits: 0 missing=191 recoE=0.000000
subgraph simHits: 191 simE=0.057819
subgraph recHits: 0 missing=191 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (22.593, 53.439, -297.945, 0.000)
p4@checkpoint: (0.774, 1.832, -10.284, 10.475)
raw GEN: n/a
raw SIM: SimTrack #454 key=424969

e- (11)
Particle 83
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (9.704, 3.176, -28.825, 30.580)
nProdVtx: 1 nDecayVtx: 7
nParents: 1 nChildren: 7
nCheckpoints: 1
direct simHits: 152 simE=0.074799
direct recHits: 0 missing=152 recoE=0.000000
subgraph simHits: 460 simE=0.266239
subgraph recHits: 0 missing=460 recoE=0.000000
direct tracker simHits: 19 dE=19.528147
subgraph tracker simHits: 16 dE=19.528143
checkpointId: 0
x4@checkpoint: (96.020, 40.984, -293.500, 0.000)
p4@checkpoint: (2.775, 1.718, -9.222, 9.783)
raw GEN: n/a
raw SIM: SimTrack #455 key=448364

Bremsstrahlung
Vertex 43
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (25.195, 8.307, -72.759, 2.399)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #154 key=82

Bremsstrahlung
Vertex 44
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (32.385, 10.788, -94.258, 3.160)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #155 key=83

Bremsstrahlung
Vertex 45
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (51.694, 18.199, -152.668, 5.227)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #156 key=84

Bremsstrahlung
Vertex 46
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (53.224, 18.843, -157.354, 5.393)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #157 key=85

Bremsstrahlung
Vertex 47
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (61.506, 22.494, -182.890, 6.296)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #158 key=86

Bremsstrahlung
Vertex 48
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (74.781, 28.956, -224.561, 7.771)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #160 key=88

Bremsstrahlung
Vertex 49
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (87.627, 35.967, -265.872, 9.233)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #162 key=90

e+ (-11)
Particle 84
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (3.276, 1.072, -9.730, 10.322)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 239 simE=0.087414
subgraph recHits: 0 missing=239 recoE=0.000000
direct tracker simHits: 26 dE=24.944561
subgraph tracker simHits: 17 dE=36.604759
raw GEN: n/a
raw SIM: SimTrack #456 key=448365

Bremsstrahlung
Vertex 38
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (25.208, 8.068, -72.649, 2.395)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #141 key=69

Bremsstrahlung
Vertex 39
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (31.103, 4.965, -229.445, 8.019)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #144 key=72

gamma (22)
Particle 85
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (3.251, 0.939, -9.586, 10.165)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 238 simE=0.087355
subgraph recHits: 0 missing=238 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 7 dE=11.660198
raw GEN: n/a
raw SIM: SimTrack #457 key=448366

PairConversion

Vertex 40
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (62.609, 18.865, -182.921, 6.296)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #150 key=78

gamma (22)
Particle 86
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.011, -0.022, 0.024)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000060
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000060
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (34.230, -28.304, -293.500, 0.000)
p4@checkpoint: (0.001, -0.011, -0.022, 0.024)
raw GEN: n/a
raw SIM: SimTrack #460 key=448378

e- (11)
Particle 87
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.615, 0.466, -4.759, 5.047)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 120 simE=0.045985
direct recHits: 0 missing=120 recoE=0.000000
subgraph simHits: 121 simE=0.046095
subgraph recHits: 0 missing=121 recoE=0.000000
direct tracker simHits: 6 dE=5.758708
subgraph tracker simHits: 5 dE=5.758708
checkpointId: 0
x4@checkpoint: (98.465, 34.375, -293.500, 0.000)
p4@checkpoint: (1.451, 0.854, -4.751, 5.041)
raw GEN: n/a
raw SIM: SimTrack #467 key=448473

Bremsstrahlung
Vertex 42
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (90.543, 30.008, -267.973, 9.305)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #153 key=81

e+ (-11)
Particle 88
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.637, 0.473, -4.827, 5.118)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 82 simE=0.030298
direct recHits: 0 missing=82 recoE=0.000000
subgraph simHits: 117 simE=0.041260
subgraph recHits: 0 missing=117 recoE=0.000000
direct tracker simHits: 6 dE=5.901490
subgraph tracker simHits: 5 dE=5.901490
checkpointId: 0
x4@checkpoint: (101.039, 24.048, -293.500, 0.000)
p4@checkpoint: (1.191, -0.039, -3.385, 3.588)
raw GEN: n/a
raw SIM: SimTrack #468 key=448474

Bremsstrahlung
Vertex 41
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (75.568, 22.002, -220.770, 7.634)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #151 key=79

gamma (22)
Particle 89
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.498, 0.099, -1.441, 1.527)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 35 simE=0.010962
direct recHits: 0 missing=35 recoE=0.000000
subgraph simHits: 35 simE=0.010962
subgraph recHits: 0 missing=35 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.689, 26.981, -293.500, 0.000)
p4@checkpoint: (0.498, 0.099, -1.441, 1.527)
raw GEN: n/a
raw SIM: SimTrack #469 key=448475

gamma (22)
Particle 90
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, -0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000110
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000110
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.588, 34.147, -293.500, 0.000)
p4@checkpoint: (0.001, 0.000, -0.003, 0.003)
raw GEN: n/a
raw SIM: SimTrack #471 key=456990

gamma (22)
Particle 91
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.124, 0.042, -0.369, 0.391)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 20 simE=0.004580
direct recHits: 0 missing=20 recoE=0.000000
subgraph simHits: 20 simE=0.004580
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (99.128, 33.468, -293.500, 0.000)
p4@checkpoint: (0.124, 0.042, -0.369, 0.391)
raw GEN: n/a
raw SIM: SimTrack #472 key=465450

gamma (22)
Particle 92
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (5.767, 2.020, -17.268, 18.317)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 249 simE=0.168285
direct recHits: 0 missing=249 recoE=0.000000
subgraph simHits: 249 simE=0.168285
subgraph recHits: 0 missing=249 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.930, 34.098, -293.500, 0.000)
p4@checkpoint: (5.767, 2.020, -17.268, 18.317)
raw GEN: n/a
raw SIM: SimTrack #473 key=465451

gamma (22)
Particle 93
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.248, 0.104, -0.760, 0.806)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 25 simE=0.006202
direct recHits: 0 missing=25 recoE=0.000000
subgraph simHits: 25 simE=0.006202
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.740, 37.439, -293.500, 0.000)
p4@checkpoint: (0.248, 0.104, -0.760, 0.806)
raw GEN: n/a
raw SIM: SimTrack #474 key=465453

gamma (22)
Particle 94
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.183, 0.078, -0.561, 0.595)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 24 simE=0.007538
direct recHits: 0 missing=24 recoE=0.000000
subgraph simHits: 24 simE=0.007538
subgraph recHits: 0 missing=24 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.619, 37.689, -293.500, 0.000)
p4@checkpoint: (0.183, 0.078, -0.561, 0.595)
raw GEN: n/a
raw SIM: SimTrack #475 key=465455

gamma (22)
Particle 95
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.097, 0.045, -0.302, 0.320)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.002387
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.002387
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (97.156, 38.818, -293.500, 0.000)
p4@checkpoint: (0.097, 0.045, -0.302, 0.320)
raw GEN: n/a
raw SIM: SimTrack #476 key=465456

gamma (22)
Particle 96
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.022, 0.011, -0.070, 0.074)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000436
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000436
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.489, 40.156, -293.500, 0.000)
p4@checkpoint: (0.022, 0.011, -0.070, 0.074)
raw GEN: n/a
raw SIM: SimTrack #478 key=465458

gamma (22)
Particle 97
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.076, 0.044, -0.246, 0.261)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.002013
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.002013
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (96.107, 40.852, -293.500, 0.000)
p4@checkpoint: (0.076, 0.044, -0.246, 0.261)
raw GEN: n/a
raw SIM: SimTrack #480 key=465460

e- (11)
Particle 98
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (56.797, 18.741, -169.582, 179.820)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1017 simE=1.194491
direct recHits: 0 missing=1017 recoE=0.000000
subgraph simHits: 1017 simE=1.194491
subgraph recHits: 0 missing=1017 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.685, 36.193, -325.332, 0.000)
p4@checkpoint: (55.790, 18.409, -166.580, 176.636)
raw GEN: n/a
raw SIM: SimTrack #482 key=517199

e+ (-11)
Particle 99
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.023, 0.007, -0.065, 0.069)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000215
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000215
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.686, 36.193, -325.333, 0.000)
p4@checkpoint: (0.022, 0.008, -0.065, 0.069)
raw GEN: n/a
raw SIM: SimTrack #483 key=517200

e- (11)
Particle 100
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.313, 0.791, 0.731, 1.122)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 45 simE=0.754643
subgraph recHits: 0 missing=45 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #484 key=825249

Bremsstrahlung
Vertex 56
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.666, 107.673, 101.677, 4.841)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #170 key=98

Bremsstrahlung
Vertex 57
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.683, 107.716, 101.717, 4.843)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #171 key=99

Bremsstrahlung
Vertex 58
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.872, 108.211, 102.176, 4.867)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #172 key=100

Bremsstrahlung
Vertex 59
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.902, 108.291, 102.250, 4.870)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #173 key=101

Bremsstrahlung
Vertex 60
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.939, 108.391, 102.343, 4.875)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #174 key=102

Bremsstrahlung
Vertex 61
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.949, 108.419, 102.368, 4.876)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #175 key=103

Bremsstrahlung
Vertex 62
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.961, 108.455, 102.399, 4.878)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #176 key=104

Bremsstrahlung
Vertex 63
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.111, 109.555, 103.417, 4.929)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #177 key=105

Bremsstrahlung
Vertex 64
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.111, 109.559, 103.420, 4.929)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #178 key=106

e+ (-11)
Particle 101
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (3.348, 8.447, 7.808, 11.980)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 150 simE=4.483890
direct recHits: 0 missing=150 recoE=0.000000
subgraph simHits: 189 simE=8.580088
subgraph recHits: 0 missing=189 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.470, 114.610, 108.107, 0.000)
p4@checkpoint: (1.981, 4.796, 4.459, 6.842)
raw GEN: n/a
raw SIM: SimTrack #485 key=825250

Bremsstrahlung
Vertex 52
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.877, 108.202, 102.166, 4.866)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #166 key=94

Bremsstrahlung
Vertex 53
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.124, 108.822, 102.740, 4.896)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #167 key=95

Bremsstrahlung
Vertex 54
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.947, 110.877, 104.643, 4.993)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #168 key=96

Bremsstrahlung
Vertex 55
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (43.953, 110.892, 104.656, 4.994)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #169 key=97

gamma (22)
Particle 102
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.253, 0.637, 0.589, 0.903)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 38 simE=0.691030
direct recHits: 0 missing=38 recoE=0.000000
subgraph simHits: 38 simE=0.691030
subgraph recHits: 0 missing=38 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.427, 114.627, 108.106, 0.000)
p4@checkpoint: (0.253, 0.637, 0.589, 0.903)
raw GEN: n/a
raw SIM: SimTrack #486 key=825251

gamma (22)
Particle 103
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.053, 2.643, 2.445, 3.751)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 68 simE=3.150058
direct recHits: 0 missing=68 recoE=0.000000
subgraph simHits: 68 simE=3.150058
subgraph recHits: 0 missing=68 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.434, 114.624, 108.106, 0.000)
p4@checkpoint: (1.053, 2.643, 2.445, 3.751)
raw GEN: n/a
raw SIM: SimTrack #487 key=825252

gamma (22)
Particle 104
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000232
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000232
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.454, 114.616, 108.106, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #488 key=825254

gamma (22)
Particle 105
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.135, 0.335, 0.310, 0.476)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 38 simE=0.254881
direct recHits: 0 missing=38 recoE=0.000000
subgraph simHits: 38 simE=0.254881
subgraph recHits: 0 missing=38 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.454, 114.616, 108.107, 0.000)
p4@checkpoint: (0.135, 0.335, 0.310, 0.476)
raw GEN: n/a
raw SIM: SimTrack #489 key=825255

gamma (22)
Particle 106
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.040, 0.103, 0.096, 0.146)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.103380
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.103380
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.359, 114.654, 108.180, 0.000)
p4@checkpoint: (0.040, 0.103, 0.096, 0.146)
raw GEN: n/a
raw SIM: SimTrack #490 key=832298

gamma (22)
Particle 107
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.008, 0.007, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003967
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003967
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.057, 114.772, 115.096, 0.000)
p4@checkpoint: (0.001, 0.002, 0.004, 0.004)
raw GEN: n/a
raw SIM: SimTrack #491 key=832299

gamma (22)
Particle 108
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004034
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004034
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.307, 114.674, 108.176, 0.000)
p4@checkpoint: (0.001, 0.003, 0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #492 key=832300

gamma (22)
Particle 109
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.020, 0.052, 0.048, 0.074)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 12 simE=0.072042
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 12 simE=0.072042
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.286, 114.682, 108.150, 0.000)
p4@checkpoint: (0.020, 0.052, 0.048, 0.074)
raw GEN: n/a
raw SIM: SimTrack #493 key=832301

gamma (22)
Particle 110
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000281
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000281
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.247, 114.698, 108.120, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #494 key=832302

gamma (22)
Particle 111
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.216, 0.593, 0.542, 0.831)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 37 simE=0.544501
direct recHits: 0 missing=37 recoE=0.000000
subgraph simHits: 37 simE=0.544501
subgraph recHits: 0 missing=37 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.239, 114.701, 108.110, 0.000)
p4@checkpoint: (0.216, 0.593, 0.542, 0.831)
raw GEN: n/a
raw SIM: SimTrack #495 key=832303

gamma (22)
Particle 112
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000386
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000386
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.036, 114.781, 108.006, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #496 key=832305

gamma (22)
Particle 113
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000087
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000087
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (42.592, 115.710, 108.859, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #497 key=832307

gamma (22)
Particle 114
caloBoundary
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.020, 0.018, 0.027)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.025965
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.025965
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (42.642, 115.692, 108.852, 0.000)
p4@checkpoint: (-0.002, 0.020, 0.018, 0.027)
raw GEN: n/a
raw SIM: SimTrack #498 key=832308

e- (11)
Particle 115
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (21.769, 55.029, 50.485, 77.786)
nProdVtx: 1 nDecayVtx: 18
nParents: 1 nChildren: 19
nCheckpoints: 1
direct simHits: 73 simE=3.701471
direct recHits: 0 missing=73 recoE=0.000000
subgraph simHits: 401 simE=70.298279
subgraph recHits: 0 missing=401 recoE=0.000000
direct tracker simHits: 23 dE=23.306692
subgraph tracker simHits: 36 dE=89.639084
checkpointId: 0
x4@checkpoint: (44.051, 115.163, 107.326, 0.000)
p4@checkpoint: (0.966, 2.867, 2.573, 3.972)
raw GEN: n/a
raw SIM: SimTrack #499 key=832976

Bremsstrahlung
Vertex 113
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (3.944, 9.968, 11.309, 0.218)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #238 key=166

Bremsstrahlung
Vertex 114
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.468, 13.833, 14.853, 0.400)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #239 key=167

Bremsstrahlung
Vertex 115
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.697, 19.502, 20.050, 0.667)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #240 key=168

Bremsstrahlung
Vertex 116
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (24.331, 62.423, 59.316, 2.685)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #241 key=169

Bremsstrahlung
Vertex 117
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.470, 107.864, 100.738, 4.814)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #242 key=170

Bremsstrahlung
Vertex 118
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.482, 107.897, 100.768, 4.816)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #243 key=171

Bremsstrahlung
Vertex 119
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.485, 107.904, 100.775, 4.816)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #244 key=172

Bremsstrahlung
Vertex 120
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.532, 108.030, 100.889, 4.822)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #245 key=173

Bremsstrahlung
Vertex 121
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.532, 108.031, 100.890, 4.822)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #246 key=174

Bremsstrahlung
Vertex 122
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.541, 108.055, 100.912, 4.823)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #247 key=175

Bremsstrahlung
Vertex 123
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.547, 108.071, 100.927, 4.824)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #248 key=176

Bremsstrahlung
Vertex 124
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.557, 108.097, 100.950, 4.825)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #249 key=177

Bremsstrahlung
Vertex 125
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.563, 108.113, 100.965, 4.826)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #250 key=178

Bremsstrahlung
Vertex 126
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (28.010, 112.268, 117.259, 6.350)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #251 key=179

Bremsstrahlung
Vertex 127
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (112.704, -27.196, 185.507, 11.521)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #252 key=180

Annihilation

Vertex 128
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (112.790, -27.228, 185.522, 11.523)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #253 key=181

Annihilation

Vertex 133
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.856, 108.230, 101.243, 4.885)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #259 key=187

Bremsstrahlung
Vertex 134
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.769, 108.444, 101.282, 4.843)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #260 key=188

e+ (-11)
Particle 116
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (8.574, 21.673, 19.883, 30.636)
nProdVtx: 1 nDecayVtx: 11
nParents: 1 nChildren: 11
nCheckpoints: 1
direct simHits: 43 simE=1.108285
direct recHits: 0 missing=43 recoE=0.000000
subgraph simHits: 295 simE=27.939732
subgraph recHits: 0 missing=295 recoE=0.000000
direct tracker simHits: 24 dE=24.496231
subgraph tracker simHits: 14 dE=24.496231
checkpointId: 0
x4@checkpoint: (49.455, 112.947, 107.401, 0.000)
p4@checkpoint: (0.503, 0.870, 0.856, 1.320)
raw GEN: n/a
raw SIM: SimTrack #500 key=832977

Bremsstrahlung
Vertex 66
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.485, 13.820, 14.848, 0.400)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #180 key=108

Bremsstrahlung
Vertex 67
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (14.797, 36.625, 35.861, 1.480)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #181 key=109

Bremsstrahlung
Vertex 68
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (33.792, 80.471, 76.626, 3.575)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #182 key=110

Bremsstrahlung
Vertex 69
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (33.872, 80.648, 76.792, 3.583)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #183 key=111

Bremsstrahlung
Vertex 70
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.808, 105.995, 100.703, 4.812)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #184 key=112

Bremsstrahlung
Vertex 71
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.891, 106.164, 100.863, 4.820)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #185 key=113

Bremsstrahlung
Vertex 72
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.902, 106.186, 100.884, 4.821)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #186 key=114

Bremsstrahlung
Vertex 73
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.920, 106.222, 100.919, 4.823)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #187 key=115

Ionisation

Vertex 74
domain: SIM
reason: Ionisation

hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.189, 106.767, 101.437, 4.850)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #188 key=116

Bremsstrahlung
Vertex 75
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.306, 107.001, 101.662, 4.861)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #189 key=117

Bremsstrahlung
Vertex 76
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.081, 108.552, 103.145, 4.937)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #190 key=118

gamma (22)
Particle 117
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.421, 3.552, 3.264, 5.029)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 104 simE=4.679423
direct recHits: 0 missing=104 recoE=0.000000
subgraph simHits: 104 simE=4.679423
subgraph recHits: 0 missing=104 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.756, 114.496, 107.352, 0.000)
p4@checkpoint: (1.421, 3.552, 3.264, 5.029)
raw GEN: n/a
raw SIM: SimTrack #501 key=832978

gamma (22)
Particle 118
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000424
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000424
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.029, 113.979, 107.315, 0.000)
p4@checkpoint: (0.001, 0.003, 0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #502 key=832979

gamma (22)
Particle 119
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.166, 0.369, 0.345, 0.532)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 40 simE=0.312471
subgraph recHits: 0 missing=40 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #503 key=832981

PairConversion

Vertex 91
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.370, 106.199, 100.699, 4.812)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #205 key=133

gamma (22)
Particle 120
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (4.011, 8.913, 8.341, 12.849)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 155 simE=11.931522
subgraph recHits: 0 missing=155 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #504 key=832982

PairConversion

Vertex 84
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.464, 106.408, 100.896, 4.822)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #198 key=126

gamma (22)
Particle 121
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.307, 0.623, 0.593, 0.914)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 62 simE=0.719693
subgraph recHits: 0 missing=62 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #505 key=832986

PairConversion

Vertex 77
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.889, 106.161, 100.860, 4.820)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #191 key=119

gamma (22)
Particle 122
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, 0.013, 0.013, 0.020)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.019195
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.019195
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.281, 113.023, 107.389, 0.000)
p4@checkpoint: (0.007, 0.013, 0.013, 0.020)
raw GEN: n/a
raw SIM: SimTrack #506 key=832987

gamma (22)
Particle 123
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000289
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000289
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (43.885, 115.226, 106.432, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #507 key=832988

gamma (22)
Particle 124
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001037
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001037
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.283, 113.022, 107.389, 0.000)
p4@checkpoint: (0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #508 key=832989

e- (11)
Particle 125
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.153, 0.353, 0.314, 0.496)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 22 simE=0.472802
direct recHits: 0 missing=22 recoE=0.000000
subgraph simHits: 22 simE=0.472802
subgraph recHits: 0 missing=22 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.336, 113.430, 107.076, 0.000)
p4@checkpoint: (0.080, 0.375, 0.307, 0.491)
raw GEN: n/a
raw SIM: SimTrack #509 key=832990

gamma (22)
Particle 126
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.335, 0.672, 0.643, 0.989)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 44 simE=0.912348
direct recHits: 0 missing=44 recoE=0.000000
subgraph simHits: 44 simE=0.912348
subgraph recHits: 0 missing=44 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.303, 113.014, 107.409, 0.000)
p4@checkpoint: (0.335, 0.672, 0.643, 0.989)
raw GEN: n/a
raw SIM: SimTrack #510 key=832991

gamma (22)
Particle 127
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (2.878, 5.740, 5.494, 8.450)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 117 simE=7.782249
direct recHits: 0 missing=117 recoE=0.000000
subgraph simHits: 117 simE=7.782249
subgraph recHits: 0 missing=117 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.315, 113.008, 107.410, 0.000)
p4@checkpoint: (2.878, 5.740, 5.494, 8.450)
raw GEN: n/a
raw SIM: SimTrack #511 key=832992

e- (11)
Particle 128
pid: e- (11)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.056, 0.114, 0.109, 0.167)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 11 simE=0.097680
direct recHits: 0 missing=11 recoE=0.000000
subgraph simHits: 15 simE=0.105051
subgraph recHits: 0 missing=15 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (46.328, 114.266, 106.733, 0.000)
p4@checkpoint: (-0.040, 0.118, 0.087, 0.152)
raw GEN: n/a
raw SIM: SimTrack #512 key=840167

Bremsstrahlung
Vertex 81
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.243, 107.064, 101.649, 4.862)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #195 key=123

Bremsstrahlung
Vertex 82
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.338, 107.329, 101.842, 4.873)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #196 key=124

Bremsstrahlung
Vertex 83
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.840, 109.583, 103.424, 4.967)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #197 key=125

e+ (-11)
Particle 129
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.250, 0.511, 0.483, 0.746)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 22 simE=0.279342
direct recHits: 0 missing=22 recoE=0.000000
subgraph simHits: 49 simE=0.614642
subgraph recHits: 0 missing=49 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.643, 112.420, 108.010, 0.000)
p4@checkpoint: (0.172, 0.151, 0.208, 0.310)
raw GEN: n/a
raw SIM: SimTrack #513 key=840168

Bremsstrahlung
Vertex 78
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.893, 106.169, 100.868, 4.821)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #192 key=120

Bremsstrahlung
Vertex 79
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.927, 106.238, 100.933, 4.824)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #193 key=121

Bremsstrahlung
Vertex 80
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.411, 108.812, 103.599, 4.957)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #194 key=122

gamma (22)
Particle 130
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.014, 0.029, 0.027, 0.042)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.006579
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.006579
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.236, 113.043, 107.357, 0.000)
p4@checkpoint: (0.014, 0.029, 0.027, 0.042)
raw GEN: n/a
raw SIM: SimTrack #514 key=840169

gamma (22)
Particle 131
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.130, 0.260, 0.252, 0.385)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 29 simE=0.327471
direct recHits: 0 missing=29 recoE=0.000000
subgraph simHits: 29 simE=0.327471
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.315, 113.009, 107.495, 0.000)
p4@checkpoint: (0.130, 0.260, 0.252, 0.385)
raw GEN: n/a
raw SIM: SimTrack #515 key=840170

gamma (22)
Particle 132
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001251
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001251
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.072, 112.675, 107.848, 0.000)
p4@checkpoint: (0.001, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #516 key=840173

gamma (22)
Particle 133
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, 0.002, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.003498
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.003498
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.577, 113.328, 106.390, 0.000)
p4@checkpoint: (0.001, 0.003, 0.002, 0.004)
raw GEN: n/a
raw SIM: SimTrack #517 key=840635

gamma (22)
Particle 134
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003329
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003329
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.390, 113.408, 106.230, 0.000)
p4@checkpoint: (0.001, 0.003, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #518 key=840636

gamma (22)
Particle 135
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000544
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000544
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.192, 113.912, 106.409, 0.000)
p4@checkpoint: (0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #519 key=840637

e- (11)
Particle 136
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.014, 4.477, 4.188, 6.453)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 86 simE=5.356906
direct recHits: 0 missing=86 recoE=0.000000
subgraph simHits: 104 simE=6.042202
subgraph recHits: 0 missing=104 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.505, 113.358, 107.363, 0.000)
p4@checkpoint: (1.690, 3.972, 3.678, 5.671)
raw GEN: n/a
raw SIM: SimTrack #520 key=840712

Bremsstrahlung
Vertex 89
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.826, 107.215, 101.650, 4.861)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #203 key=131

e+ (-11)
Particle 137
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.997, 4.436, 4.153, 6.396)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 67 simE=2.444309
direct recHits: 0 missing=67 recoE=0.000000
subgraph simHits: 107 simE=5.889321
subgraph recHits: 0 missing=107 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.695, 113.277, 107.409, 0.000)
p4@checkpoint: (0.907, 1.831, 1.758, 2.695)
raw GEN: n/a
raw SIM: SimTrack #521 key=840713

Bremsstrahlung
Vertex 85
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.470, 106.420, 100.908, 4.823)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #199 key=127

Bremsstrahlung
Vertex 86
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.494, 106.475, 100.959, 4.825)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #200 key=128

Bremsstrahlung
Vertex 87
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.497, 106.481, 100.964, 4.826)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #201 key=129

Bremsstrahlung
Vertex 88
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.597, 108.895, 103.234, 4.942)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #202 key=130

gamma (22)
Particle 138
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, 0.003, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004796
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004796
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.576, 113.328, 107.372, 0.000)
p4@checkpoint: (0.001, 0.003, 0.003, 0.005)
raw GEN: n/a
raw SIM: SimTrack #522 key=840714

gamma (22)
Particle 139
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000211
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000211
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.155, 113.927, 111.531, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #523 key=840715

gamma (22)
Particle 140
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.002, 2.230, 2.085, 3.213)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 67 simE=3.013782
direct recHits: 0 missing=67 recoE=0.000000
subgraph simHits: 67 simE=3.013782
subgraph recHits: 0 missing=67 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.575, 113.328, 107.368, 0.000)
p4@checkpoint: (1.002, 2.230, 2.085, 3.213)
raw GEN: n/a
raw SIM: SimTrack #524 key=840716

gamma (22)
Particle 141
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.152, 0.327, 0.309, 0.475)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 25 simE=0.426221
direct recHits: 0 missing=25 recoE=0.000000
subgraph simHits: 25 simE=0.426221
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.639, 113.301, 107.392, 0.000)
p4@checkpoint: (0.152, 0.327, 0.309, 0.475)
raw GEN: n/a
raw SIM: SimTrack #525 key=840722

gamma (22)
Particle 142
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.240, 0.538, 0.502, 0.774)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 45 simE=0.684729
direct recHits: 0 missing=45 recoE=0.000000
subgraph simHits: 47 simE=0.685296
subgraph recHits: 0 missing=47 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.563, 113.334, 107.365, 0.000)
p4@checkpoint: (0.240, 0.538, 0.502, 0.774)
raw GEN: n/a
raw SIM: SimTrack #526 key=844889

Annihilation

Vertex 90
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-81.571, 74.244, 158.186, 10.705)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #204 key=132

gamma (22)
Particle 143
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000194
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000194
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.768, -24.858, 127.505, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #527 key=848935

gamma (22)
Particle 144
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000373
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000373
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-58.289, 108.652, 152.252, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #528 key=848936

e- (11)
Particle 145
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.061, 0.133, 0.125, 0.193)
nProdVtx: 1 nDecayVtx: 12
nParents: 1 nChildren: 12
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 17 simE=0.033232
subgraph recHits: 0 missing=17 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #529 key=849107

Bremsstrahlung
Vertex 101
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.455, 106.373, 100.864, 4.820)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #218 key=146

Bremsstrahlung
Vertex 102
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.485, 106.435, 100.929, 4.824)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #219 key=147

Bremsstrahlung
Vertex 103
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.666, 106.980, 101.405, 4.849)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #220 key=148

Bremsstrahlung
Vertex 104
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.696, 107.113, 101.526, 4.855)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #221 key=149

Bremsstrahlung
Vertex 105
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.704, 107.166, 101.575, 4.857)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #222 key=150

Bremsstrahlung
Vertex 106
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.304, 109.887, 105.699, 5.078)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #223 key=151

Bremsstrahlung
Vertex 107
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.681, 107.123, 101.496, 4.854)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #224 key=152

Bremsstrahlung
Vertex 108
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.158, 107.808, 104.661, 5.110)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #225 key=153

Bremsstrahlung
Vertex 109
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.583, 108.187, 103.713, 5.163)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #227 key=155

Bremsstrahlung
Vertex 110
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.652, 106.767, 101.293, 4.841)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #228 key=156

Bremsstrahlung
Vertex 111
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.123, 106.143, 102.030, 4.906)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #229 key=157

Annihilation

Vertex 112
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.014, 106.050, 102.032, 4.919)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #230 key=158

e+ (-11)
Particle 146
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.106, 0.236, 0.220, 0.340)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 27 simE=0.279239
subgraph recHits: 0 missing=27 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #530 key=849108

Bremsstrahlung
Vertex 92
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.398, 106.262, 100.757, 4.815)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #206 key=134

Bremsstrahlung
Vertex 93
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.401, 106.268, 100.763, 4.815)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #207 key=135

Bremsstrahlung
Vertex 94
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.417, 106.300, 100.792, 4.817)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #208 key=136

Bremsstrahlung
Vertex 95
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.421, 106.308, 100.799, 4.817)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #209 key=137

Bremsstrahlung
Vertex 96
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.979, 106.817, 101.563, 4.853)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #210 key=138

Bremsstrahlung
Vertex 97
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (46.173, 106.996, 101.877, 4.867)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #211 key=139

Bremsstrahlung
Vertex 98
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (48.434, 107.675, 105.412, 5.012)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #212 key=140

Bremsstrahlung
Vertex 99
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (49.510, 105.173, 110.465, 5.211)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #213 key=141

Annihilation

Vertex 100
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (48.384, 105.066, 112.928, 5.364)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #215 key=143

gamma (22)
Particle 147
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.086, 0.202, 0.185, 0.287)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.275059
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.275059
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.451, 113.382, 107.283, 0.000)
p4@checkpoint: (0.086, 0.202, 0.185, 0.287)
raw GEN: n/a
raw SIM: SimTrack #531 key=849109

gamma (22)
Particle 148
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000542
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000542
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.766, 113.246, 106.903, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #532 key=849110

gamma (22)
Particle 149
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000299
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000299
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.172, 113.071, 107.266, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #533 key=849111

gamma (22)
Particle 150
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000046
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000046
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-3.565, 123.248, 121.174, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #534 key=849112

gamma (22)
Particle 151
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, 0.002, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.002139
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.002139
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.809, 111.418, 110.283, 0.000)
p4@checkpoint: (0.001, 0.001, 0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #535 key=849113

gamma (22)
Particle 152
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000267
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000267
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.873, 112.316, 110.652, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #536 key=849116

gamma (22)
Particle 153
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000238
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000238
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.947, 112.282, 111.703, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #537 key=849118

gamma (22)
Particle 154
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000254
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000254
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.699, 112.840, 127.612, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #538 key=849119

gamma (22)
Particle 155
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000397
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000397
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-71.183, -100.677, 215.372, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #540 key=849134

gamma (22)
Particle 156
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.017, 0.034, 0.036, 0.052)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.002057
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.002057
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.765, 113.247, 107.977, 0.000)
p4@checkpoint: (0.017, 0.034, 0.036, 0.052)
raw GEN: n/a
raw SIM: SimTrack #543 key=849332

gamma (22)
Particle 157
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000850
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000850
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.783, 113.239, 107.211, 0.000)
p4@checkpoint: (0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #544 key=849334

gamma (22)
Particle 158
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003234
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003234
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.424, 113.815, 107.728, 0.000)
p4@checkpoint: (0.001, 0.003, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #545 key=849338

gamma (22)
Particle 159
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000082
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000082
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (46.981, 113.999, 107.678, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #546 key=849340

gamma (22)
Particle 160
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000050
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000050
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (46.639, 114.139, 108.307, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #547 key=849342

gamma (22)
Particle 161
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.010, -0.005, 0.010, 0.015)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.014974
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.014974
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.102, 31.900, 260.224, 0.000)
p4@checkpoint: (-0.010, -0.005, 0.010, 0.015)
raw GEN: n/a
raw SIM: SimTrack #548 key=849346

gamma (22)
Particle 162
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.004, 0.003, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004645
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004645
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (46.417, 114.229, 106.825, 0.000)
p4@checkpoint: (0.000, 0.004, 0.003, 0.005)
raw GEN: n/a
raw SIM: SimTrack #549 key=849368

gamma (22)
Particle 163
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.002, -0.002, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.005258
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.005258
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.840, 112.331, 100.137, 0.000)
p4@checkpoint: (0.004, 0.002, -0.002, 0.005)
raw GEN: n/a
raw SIM: SimTrack #550 key=849378

gamma (22)
Particle 164
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000661
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000661
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-109.151, 57.349, 87.386, 0.000)
p4@checkpoint: (-0.001, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #552 key=849383

gamma (22)
Particle 165
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000833
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000833
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.710, 112.835, 108.919, 0.000)
p4@checkpoint: (0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #553 key=849397

gamma (22)
Particle 166
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000129
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000129
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (41.179, 116.220, 102.514, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #554 key=849402

gamma (22)
Particle 167
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000457
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000457
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (59.324, 108.091, 87.418, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #555 key=849405

gamma (22)
Particle 168
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, 0.017, 0.016, 0.024)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.018545
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.018545
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 2 dE=2.088261
subgraph tracker simHits: 2 dE=2.088261
checkpointId: 0
x4@checkpoint: (40.539, 116.445, 122.694, 0.000)
p4@checkpoint: (0.004, 0.013, 0.014, 0.019)
raw GEN: n/a
raw SIM: SimTrack #563 key=852656

gamma (22)
Particle 169
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.039, 0.099, 0.091, 0.140)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.014503
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 60 dE=58.313416
subgraph tracker simHits: 22 dE=62.127453
raw GEN: n/a
raw SIM: SimTrack #564 key=852657

PairConversion

Vertex 135
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.803, 19.762, 20.290, 0.679)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #261 key=189

gamma (22)
Particle 170
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.003, 0.003, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004600
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004600
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.089, 114.760, 107.356, 0.000)
p4@checkpoint: (0.001, 0.003, 0.003, 0.005)
raw GEN: n/a
raw SIM: SimTrack #565 key=852658

gamma (22)
Particle 171
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.031, 0.080, 0.073, 0.113)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.106724
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.106724
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.447, 115.010, 107.338, 0.000)
p4@checkpoint: (0.031, 0.080, 0.073, 0.113)
raw GEN: n/a
raw SIM: SimTrack #566 key=852659

gamma (22)
Particle 172
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.307, 0.826, 0.752, 1.159)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 37 simE=1.053489
direct recHits: 0 missing=37 recoE=0.000000
subgraph simHits: 37 simE=1.053489
subgraph recHits: 0 missing=37 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.167, 115.118, 107.339, 0.000)
p4@checkpoint: (0.307, 0.826, 0.752, 1.159)
raw GEN: n/a
raw SIM: SimTrack #567 key=852663

gamma (22)
Particle 173
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (9.672, 26.012, 23.675, 36.478)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 196 simE=31.979143
direct recHits: 0 missing=196 recoE=0.000000
subgraph simHits: 196 simE=31.979143
subgraph recHits: 0 missing=196 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.167, 115.118, 107.340, 0.000)
p4@checkpoint: (9.672, 26.012, 23.675, 36.478)
raw GEN: n/a
raw SIM: SimTrack #568 key=852664

gamma (22)
Particle 174
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.051, 0.136, 0.124, 0.191)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.182366
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.182366
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.167, 115.118, 107.340, 0.000)
p4@checkpoint: (0.051, 0.136, 0.124, 0.191)
raw GEN: n/a
raw SIM: SimTrack #569 key=852665

gamma (22)
Particle 175
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (9.347, 25.132, 22.868, 35.241)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 289 simE=32.843178
subgraph recHits: 0 missing=289 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #570 key=852666

PairConversion

Vertex 129
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.226, 109.895, 102.586, 4.909)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #254 key=182

gamma (22)
Particle 176
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.034, 0.092, 0.084, 0.129)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.124992
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.124992
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.170, 115.117, 107.345, 0.000)
p4@checkpoint: (0.034, 0.092, 0.084, 0.129)
raw GEN: n/a
raw SIM: SimTrack #571 key=852667

gamma (22)
Particle 177
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001200
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001200
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.172, 115.116, 107.344, 0.000)
p4@checkpoint: (0.000, 0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #572 key=852668

gamma (22)
Particle 178
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.041, 0.110, 0.101, 0.155)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.143367
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.143367
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.169, 115.117, 107.349, 0.000)
p4@checkpoint: (0.041, 0.110, 0.101, 0.155)
raw GEN: n/a
raw SIM: SimTrack #573 key=852669

gamma (22)
Particle 179
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, 0.016, 0.015, 0.023)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.022781
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.022781
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.167, 115.118, 107.343, 0.000)
p4@checkpoint: (0.006, 0.016, 0.015, 0.023)
raw GEN: n/a
raw SIM: SimTrack #574 key=852670

gamma (22)
Particle 180
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.026, 0.070, 0.063, 0.098)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.096170
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.096170
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.165, 115.119, 107.340, 0.000)
p4@checkpoint: (0.026, 0.070, 0.063, 0.098)
raw GEN: n/a
raw SIM: SimTrack #575 key=852671

gamma (22)
Particle 181
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=2.116691
subgraph tracker simHits: 2 dE=2.116691
checkpointId: 0
x4@checkpoint: (-4.740, -114.886, 293.500, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #576 key=853122

gamma (22)
Particle 182
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.002, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.003040
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.003040
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (118.514, -34.018, 181.415, 0.000)
p4@checkpoint: (0.002, -0.002, -0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #577 key=853727

gamma (22)
Particle 183
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000100
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000100
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.800, -11.097, 186.200, 0.000)
p4@checkpoint: (0.001, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #578 key=853728

gamma (22)
Particle 184
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001819
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001819
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.757, -29.346, 189.114, 0.000)
p4@checkpoint: (0.002, -0.000, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #579 key=853729

e- (11)
Particle 185
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.213, 3.265, 2.965, 4.574)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 81 simE=4.373494
direct recHits: 0 missing=81 recoE=0.000000
subgraph simHits: 82 simE=4.373732
subgraph recHits: 0 missing=82 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.119, 115.137, 107.333, 0.000)
p4@checkpoint: (1.152, 3.283, 2.963, 4.570)
raw GEN: n/a
raw SIM: SimTrack #580 key=855510

Bremsstrahlung
Vertex 132
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (34.075, 112.724, 119.631, 6.108)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #258 key=186

e+ (-11)
Particle 186
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (8.134, 21.870, 19.900, 30.667)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 260 simE=28.468040
direct recHits: 0 missing=260 recoE=0.000000
subgraph simHits: 263 simE=28.469454
subgraph recHits: 0 missing=263 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.176, 115.115, 107.339, 0.000)
p4@checkpoint: (8.195, 21.842, 19.900, 30.664)
raw GEN: n/a
raw SIM: SimTrack #581 key=855511

Annihilation

Vertex 130
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (40.377, -102.574, -56.290, 14.890)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #256 key=184

Annihilation

Vertex 131
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (26.804, 113.197, 122.172, 6.496)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #257 key=185

gamma (22)
Particle 187
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000392
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000392
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (109.205, 57.246, -31.767, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #583 key=872792

gamma (22)
Particle 188
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (38.428, -117.159, -54.842, 0.000)
p4@checkpoint: (-0.000, -0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #584 key=872793

gamma (22)
Particle 189
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.279, 115.075, 128.513, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #585 key=874620

gamma (22)
Particle 190
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000238
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000238
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (29.611, 119.692, 127.812, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #586 key=876120

gamma (22)
Particle 191
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (44.688, 114.917, 95.614, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #587 key=901647

gamma (22)
Particle 192
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000355
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000355
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (43.732, 115.284, 106.460, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #588 key=901649

e- (11)
Particle 193
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.022, 0.057, 0.053, 0.081)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.005842
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 2 dE=1.019084
subgraph tracker simHits: 2 dE=1.019084
raw GEN: n/a
raw SIM: SimTrack #589 key=901751

Bremsstrahlung
Vertex 138
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.048, 22.769, 22.970, 0.815)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #264 key=192

e+ (-11)
Particle 194
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.017, 0.042, 0.038, 0.059)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.008661
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 6 dE=2.794945
subgraph tracker simHits: 3 dE=2.794945
raw GEN: n/a
raw SIM: SimTrack #590 key=901752

Bremsstrahlung
Vertex 136
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (11.561, 22.171, 62.649, 2.789)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #262 key=190

Annihilation

Vertex 137
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (11.581, 18.953, 145.363, 7.221)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #263 key=191

gamma (22)
Particle 195
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, 0.002, 0.005, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.008312
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.008312
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (112.477, 50.515, 126.096, 0.000)
p4@checkpoint: (0.007, 0.002, 0.005, 0.009)
raw GEN: n/a
raw SIM: SimTrack #591 key=901757

gamma (22)
Particle 196
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000349
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000349
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-4.626, 123.213, 68.077, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #592 key=901765

gamma (22)
Particle 197
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.004, 0.004, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.005842
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.005842
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-14.644, 122.427, 107.898, 0.000)
p4@checkpoint: (-0.001, 0.004, 0.004, 0.006)
raw GEN: n/a
raw SIM: SimTrack #593 key=901777

e- (11)
Particle 198
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.737, 0.976, 3.665, 3.863)
nProdVtx: 1 nDecayVtx: 4
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 12 simE=0.002761
direct recHits: 0 missing=12 recoE=0.000000
subgraph simHits: 102 simE=0.024725
subgraph recHits: 0 missing=102 recoE=0.000000
direct tracker simHits: 19 dE=19.100969
subgraph tracker simHits: 15 dE=19.100971
checkpointId: 0
x4@checkpoint: (-63.432, 24.200, 293.500, 0.000)
p4@checkpoint: (0.055, -0.123, 0.448, 0.468)
raw GEN: n/a
raw SIM: SimTrack #597 key=901825

Bremsstrahlung
Vertex 141
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-24.003, 30.116, 117.619, 3.808)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #274 key=202

Bremsstrahlung
Vertex 142
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-24.979, 31.077, 121.712, 3.952)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #275 key=203

Bremsstrahlung
Vertex 143
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-43.886, 42.083, 188.230, 6.290)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #277 key=205

Bremsstrahlung
Vertex 144
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-54.532, 42.150, 221.106, 7.444)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #279 key=207

e+ (-11)
Particle 199
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.930, 1.233, 4.632, 4.883)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 93 simE=0.040758
direct recHits: 0 missing=93 recoE=0.000000
subgraph simHits: 94 simE=0.040848
subgraph recHits: 0 missing=94 recoE=0.000000
direct tracker simHits: 40 dE=40.354752
subgraph tracker simHits: 23 dE=40.354752
checkpointId: 0
x4@checkpoint: (-41.789, 86.170, 293.500, 0.000)
p4@checkpoint: (-0.194, 1.497, 4.532, 4.777)
raw GEN: n/a
raw SIM: SimTrack #598 key=901826

Annihilation

Vertex 140
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-20.772, 28.235, 107.255, 3.443)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #273 key=201

gamma (22)
Particle 200
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000090
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000090
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.645, -12.694, 99.098, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #603 key=910013

gamma (22)
Particle 201
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.270, 0.271, 1.143, 1.205)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 31 simE=0.005539
direct recHits: 0 missing=31 recoE=0.000000
subgraph simHits: 31 simE=0.005539
subgraph recHits: 0 missing=31 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-65.580, 71.783, 293.500, 0.000)
p4@checkpoint: (-0.270, 0.271, 1.143, 1.205)
raw GEN: n/a
raw SIM: SimTrack #604 key=910015

gamma (22)
Particle 202
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.236, 0.228, 0.978, 1.031)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 31 simE=0.007068
direct recHits: 0 missing=31 recoE=0.000000
subgraph simHits: 31 simE=0.007068
subgraph recHits: 0 missing=31 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-66.392, 71.071, 293.500, 0.000)
p4@checkpoint: (-0.236, 0.228, 0.978, 1.031)
raw GEN: n/a
raw SIM: SimTrack #605 key=910016

gamma (22)
Particle 203
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.280, 0.082, 0.881, 0.928)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.007828
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=0.007828
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.335, 51.897, 293.500, 0.000)
p4@checkpoint: (-0.280, 0.082, 0.881, 0.928)
raw GEN: n/a
raw SIM: SimTrack #607 key=910019

gamma (22)
Particle 204
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.059, -0.017, 0.188, 0.198)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.001529
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=0.001529
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-77.254, 35.764, 293.500, 0.000)
p4@checkpoint: (-0.059, -0.017, 0.188, 0.198)
raw GEN: n/a
raw SIM: SimTrack #609 key=910022

e- (11)
Particle 205
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.574, 0.874, 3.212, 3.378)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 69 simE=0.026553
direct recHits: 0 missing=69 recoE=0.000000
subgraph simHits: 69 simE=0.026553
subgraph recHits: 0 missing=69 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-52.889, 80.532, 298.061, 0.000)
p4@checkpoint: (-0.574, 0.872, 3.211, 3.376)
raw GEN: n/a
raw SIM: SimTrack #612 key=916246

e+ (-11)
Particle 206
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.175, 0.267, 0.978, 1.028)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 34 simE=0.009579
direct recHits: 0 missing=34 recoE=0.000000
subgraph simHits: 34 simE=0.009579
subgraph recHits: 0 missing=34 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-52.897, 80.544, 298.104, 0.000)
p4@checkpoint: (-0.167, 0.253, 0.926, 0.975)
raw GEN: n/a
raw SIM: SimTrack #613 key=916247

gamma (22)
Particle 207
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.024, -0.008, 0.073, 0.078)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.001734
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 19 dE=17.825689
subgraph tracker simHits: 8 dE=18.452538
raw GEN: n/a
raw SIM: SimTrack #615 key=923733

PairConversion

Vertex 153
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-23.253, -7.909, 72.608, 2.236)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #295 key=223

gamma (22)
Particle 208
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-27.170, -9.161, 82.235, 87.091)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 700 simE=0.788048
direct recHits: 0 missing=700 recoE=0.000000
subgraph simHits: 700 simE=0.788048
subgraph recHits: 0 missing=700 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.233, -32.516, 293.500, 0.000)
p4@checkpoint: (-27.170, -9.161, 82.235, 87.091)
raw GEN: n/a
raw SIM: SimTrack #616 key=923734

gamma (22)
Particle 209
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.662, -0.223, 2.004, 2.122)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 59 simE=0.019996
direct recHits: 0 missing=59 recoE=0.000000
subgraph simHits: 59 simE=0.019996
subgraph recHits: 0 missing=59 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.249, -32.510, 293.500, 0.000)
p4@checkpoint: (-0.662, -0.223, 2.004, 2.122)
raw GEN: n/a
raw SIM: SimTrack #617 key=923735

gamma (22)
Particle 210
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.031, -0.011, 0.094, 0.100)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000241
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000241
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.240, -32.515, 293.500, 0.000)
p4@checkpoint: (-0.031, -0.011, 0.094, 0.100)
raw GEN: n/a
raw SIM: SimTrack #618 key=923736

gamma (22)
Particle 211
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.046, -0.015, 0.138, 0.146)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000552
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000552
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.240, -32.520, 293.500, 0.000)
p4@checkpoint: (-0.046, -0.015, 0.138, 0.146)
raw GEN: n/a
raw SIM: SimTrack #619 key=923737

gamma (22)
Particle 212
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.650, -0.213, 1.963, 2.079)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 37 simE=0.015254
direct recHits: 0 missing=37 recoE=0.000000
subgraph simHits: 37 simE=0.015254
subgraph recHits: 0 missing=37 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.419, -31.976, 293.500, 0.000)
p4@checkpoint: (-0.650, -0.213, 1.963, 2.079)
raw GEN: n/a
raw SIM: SimTrack #620 key=923739

gamma (22)
Particle 213
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.158, -0.376, 3.491, 3.697)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 74 simE=0.026574
direct recHits: 0 missing=74 recoE=0.000000
subgraph simHits: 74 simE=0.026574
subgraph recHits: 0 missing=74 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-96.456, -31.810, 293.500, 0.000)
p4@checkpoint: (-1.158, -0.376, 3.491, 3.697)
raw GEN: n/a
raw SIM: SimTrack #621 key=923741

e+ (-11)
Particle 214
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-0.022, -0.008, 0.067, 0.072)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 4 simE=0.001734
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 1 dE=0.626849
subgraph tracker simHits: 1 dE=0.626849
raw GEN: n/a
raw SIM: SimTrack #626 key=1181710

Bremsstrahlung
Vertex 154
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-24.906, -4.318, 154.188, 5.108)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #297 key=225

Bremsstrahlung
Vertex 155
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-25.434, -6.384, 205.453, 7.414)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #300 key=228

Annihilation

Vertex 156
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-25.155, -7.067, 260.675, 10.136)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #303 key=231

gamma (22)
Particle 215
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.002, 0.010, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000396
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000396
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (25.523, 21.635, 293.500, 0.000)
p4@checkpoint: (0.004, 0.002, 0.010, 0.011)
raw GEN: n/a
raw SIM: SimTrack #628 key=1181713

gamma (22)
Particle 216
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, -0.001, 0.006, 0.008)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000386
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000386
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (40.072, -25.932, 293.500, 0.000)
p4@checkpoint: (0.004, -0.001, 0.006, 0.008)
raw GEN: n/a
raw SIM: SimTrack #631 key=1181721

gamma (22)
Particle 217
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000440
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000440
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-114.986, 44.510, 257.400, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #634 key=1181729

gamma (22)
Particle 218
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.838, -78.795, 262.703, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #635 key=1181730

e- (11)
Particle 219
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.010, 0.008, -0.045, 0.047)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 5 dE=5.000000
subgraph tracker simHits: 5 dE=5.000000
checkpointId: 0
x4@checkpoint: (4.995, 12.726, -293.500, 0.000)
p4@checkpoint: (-0.005, 0.006, -0.026, 0.027)
raw GEN: n/a
raw SIM: SimTrack #637 key=1446422

n (2112)
Particle 220
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.380, 0.190, 0.024, 1.032)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.032088
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.032088
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.869, -18.729, -230.445, 0.000)
p4@checkpoint: (0.380, 0.190, 0.024, 1.032)
raw GEN: n/a
raw SIM: SimTrack #638 key=1549216

n (2112)
Particle 221
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.033, -0.050, -0.072, 0.944)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000087
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000087
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (115.141, -44.107, -243.313, 0.000)
p4@checkpoint: (0.018, -0.065, -0.061, 0.944)
raw GEN: n/a
raw SIM: SimTrack #639 key=1549217

gamma (22)
Particle 222
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.001, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.005173
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.005173
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.858, -24.419, -233.790, 0.000)
p4@checkpoint: (0.003, -0.001, -0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #640 key=1549218

gamma (22)
Particle 223
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.001, 0.000, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001209
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001209
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (117.490, -37.402, -228.243, 0.000)
p4@checkpoint: (0.000, -0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #641 key=1549220

n (2112)
Particle 224
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.023, 0.011, -0.028, 0.940)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000047
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000047
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (121.812, -19.097, -241.275, 0.000)
p4@checkpoint: (0.023, 0.011, -0.028, 0.940)
raw GEN: n/a
raw SIM: SimTrack #643 key=1549222

gamma (22)
Particle 225
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.000, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000561
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000561
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-88.157, -86.204, -226.453, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #646 key=1568270

gamma (22)
Particle 226
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000566
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000566
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-83.075, -91.112, -216.506, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #648 key=1568273

n (2112)
Particle 227
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.055, -0.412, 0.630, 1.206)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.014426
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.014426
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (30.337, -119.510, 233.668, 0.000)
p4@checkpoint: (0.099, -0.273, 0.695, 1.204)
raw GEN: n/a
raw SIM: SimTrack #651 key=1612388

gamma (22)
Particle 228
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000248
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000248
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (120.111, -27.863, -56.003, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #652 key=1612389

gamma (22)
Particle 229
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000057
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000057
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (29.308, 44.186, -293.500, 0.000)
p4@checkpoint: (0.000, 0.000, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #653 key=1651692

gamma (22)
Particle 230
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.001, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000829
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000829
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-78.127, -95.389, -38.283, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #658 key=1651716

gamma (22)
Particle 231
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000505
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000505
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-9.783, -122.911, -223.626, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #659 key=1716050

gamma (22)
Particle 232
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000553
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000553
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (1.590, -123.290, -223.168, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #660 key=1716051

n (2112)
Particle 233
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.016, -0.012, 0.022, 0.940)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (6.610, -123.123, -210.648, 0.000)
p4@checkpoint: (0.016, -0.012, 0.022, 0.940)
raw GEN: n/a
raw SIM: SimTrack #661 key=1716052

e- (11)
Particle 234
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.042, 0.122, 0.091, 0.158)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.144117
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=0.144117
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (40.712, 116.385, 104.046, 0.000)
p4@checkpoint: (-0.005, 0.128, 0.089, 0.156)
raw GEN: n/a
raw SIM: SimTrack #662 key=1760875

gamma (22)
Particle 235
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, -0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000734
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000734
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (104.779, 64.994, 114.068, 0.000)
p4@checkpoint: (0.001, -0.000, -0.002, 0.002)
raw GEN: n/a
raw SIM: SimTrack #663 key=1761097

n (2112)
Particle 236
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.038, -0.050, 0.038, 0.942)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.019183
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.019183
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (99.378, 72.986, 137.880, 0.000)
p4@checkpoint: (0.034, 0.053, 0.030, 0.942)
raw GEN: n/a
raw SIM: SimTrack #664 key=1761098

gamma (22)
Particle 237
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.000, 0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000171
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000171
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (108.764, 58.080, 134.813, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #665 key=1761099

gamma (22)
Particle 238
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000341
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000341
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (102.511, 68.516, 120.309, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #666 key=1761100

n (2112)
Particle 239
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.125, -0.128, 0.417, 1.043)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 4
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 29 simE=0.038799
subgraph recHits: 0 missing=29 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #667 key=1762050

HadronInelastic

Vertex 169
domain: SIM
reason: HadronInelastic
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (90.524, 76.935, 188.771, 11.381)
nIn: 1 nOut: 4
raw GEN: n/a
raw SIM: SimVertex #324 key=252

n (2112)
Particle 240
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.243, -0.003, -0.046, 0.972)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000007
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000007
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.284, 78.254, 183.065, 0.000)
p4@checkpoint: (0.243, -0.003, -0.046, 0.972)
raw GEN: n/a
raw SIM: SimTrack #668 key=1762051

gamma (22)
Particle 241
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000200
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000200
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (88.906, 85.432, 181.449, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #669 key=1762057

gamma (22)
Particle 242
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.001, 0.002, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000316
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000316
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (99.441, 72.900, 199.398, 0.000)
p4@checkpoint: (0.002, -0.001, 0.002, 0.003)
raw GEN: n/a
raw SIM: SimTrack #670 key=1762061

n (2112)
Particle 243
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.029, 0.028, -0.042, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001715
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001715
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (91.094, 83.095, 180.535, 0.000)
p4@checkpoint: (0.044, 0.025, -0.023, 0.941)
raw GEN: n/a
raw SIM: SimTrack #672 key=1762063

gamma (22)
Particle 244
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000194
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000194
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (89.172, 85.154, 188.107, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #673 key=1762064

n (2112)
Particle 245
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.023, -0.046, -0.024, 0.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000001
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=0.000001
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.131, 16.941, 185.542, 0.000)
p4@checkpoint: (0.023, -0.046, -0.024, 0.941)
raw GEN: n/a
raw SIM: SimTrack #674 key=1762075

n (2112)
Particle 246
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.064, 0.157, 0.080, 0.958)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 13 simE=0.032549
direct recHits: 0 missing=13 recoE=0.000000
subgraph simHits: 13 simE=0.032549
subgraph recHits: 0 missing=13 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.822, 88.530, 194.705, 0.000)
p4@checkpoint: (-0.064, 0.157, 0.080, 0.958)
raw GEN: n/a
raw SIM: SimTrack #676 key=1762101

gamma (22)
Particle 247
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.003, 0.003, 0.006)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000229
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000229
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-9.905, 30.717, 293.500, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #677 key=1762103

n (2112)
Particle 248
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.072, 0.027, 0.031, 0.943)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.006021
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.006021
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (94.996, 78.605, 190.691, 0.000)
p4@checkpoint: (0.072, 0.027, 0.031, 0.943)
raw GEN: n/a
raw SIM: SimTrack #678 key=1762104

n (2112)
Particle 249
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.017, 0.006, -0.022, 0.940)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (95.006, 78.592, 182.704, 0.000)
p4@checkpoint: (0.017, 0.006, -0.022, 0.940)
raw GEN: n/a
raw SIM: SimTrack #680 key=1762107

gamma (22)
Particle 250
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.180, -0.429, -0.397, 0.612)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 30 simE=0.596421
direct recHits: 0 missing=30 recoE=0.000000
subgraph simHits: 30 simE=0.596421
subgraph recHits: 0 missing=30 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-47.484, -113.790, -103.224, 0.000)
p4@checkpoint: (-0.180, -0.429, -0.397, 0.612)
raw GEN: n/a
raw SIM: SimTrack #681 key=1765943

gamma (22)
Particle 251
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.026, -0.061, -0.056, 0.087)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=0.086847
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=0.086847
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-47.674, -113.711, -103.229, 0.000)
p4@checkpoint: (-0.026, -0.061, -0.056, 0.087)
raw GEN: n/a
raw SIM: SimTrack #682 key=1765945

gamma (22)
Particle 252
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, -0.009, -0.008, 0.013)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.004038
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.004038
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-49.782, -112.804, -107.780, 0.000)
p4@checkpoint: (-0.002, -0.001, -0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #683 key=1765946

gamma (22)
Particle 253
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.176, -0.410, -0.381, 0.587)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 22 simE=0.582210
direct recHits: 0 missing=22 recoE=0.000000
subgraph simHits: 22 simE=0.582210
subgraph recHits: 0 missing=22 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-47.933, -113.602, -103.232, 0.000)
p4@checkpoint: (-0.176, -0.410, -0.381, 0.587)
raw GEN: n/a
raw SIM: SimTrack #684 key=1765947

gamma (22)
Particle 254
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.029, -0.068, -0.064, 0.098)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.097835
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=0.097835
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-47.933, -113.602, -103.232, 0.000)
p4@checkpoint: (-0.029, -0.068, -0.064, 0.098)
raw GEN: n/a
raw SIM: SimTrack #685 key=1765948

gamma (22)
Particle 255
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.062, -0.143, -0.133, 0.205)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.196427
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.196427
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-48.022, -113.564, -103.234, 0.000)
p4@checkpoint: (-0.062, -0.143, -0.133, 0.205)
raw GEN: n/a
raw SIM: SimTrack #686 key=1765951

gamma (22)
Particle 256
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001633
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001633
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-48.022, -113.564, -103.234, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #687 key=1765952

gamma (22)
Particle 257
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.783, -4.110, -3.831, 5.895)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 75 simE=5.817321
direct recHits: 0 missing=75 recoE=0.000000
subgraph simHits: 75 simE=5.817321
subgraph recHits: 0 missing=75 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-48.023, -113.564, -103.233, 0.000)
p4@checkpoint: (-1.783, -4.110, -3.831, 5.895)
raw GEN: n/a
raw SIM: SimTrack #688 key=1765953

gamma (22)
Particle 258
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-2.316, -5.339, -4.975, 7.656)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 88 simE=7.572259
direct recHits: 0 missing=88 recoE=0.000000
subgraph simHits: 90 simE=7.572646
subgraph recHits: 0 missing=90 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-48.022, -113.564, -103.233, 0.000)
p4@checkpoint: (-2.316, -5.339, -4.975, 7.656)
raw GEN: n/a
raw SIM: SimTrack #689 key=1765954

Bremsstrahlung
Vertex 182
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (64.865, 96.661, 5.923, 16.005)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #339 key=267

gamma (22)
Particle 259
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.192, -2.749, -2.561, 3.942)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 57 simE=3.911050
direct recHits: 0 missing=57 recoE=0.000000
subgraph simHits: 57 simE=3.911050
subgraph recHits: 0 missing=57 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-48.022, -113.564, -103.233, 0.000)
p4@checkpoint: (-1.192, -2.749, -2.561, 3.942)
raw GEN: n/a
raw SIM: SimTrack #690 key=1765955

gamma (22)
Particle 260
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-5.225, -12.019, -11.207, 17.244)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 119 simE=17.073498
direct recHits: 0 missing=119 recoE=0.000000
subgraph simHits: 119 simE=17.073498
subgraph recHits: 0 missing=119 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-48.027, -113.562, -103.234, 0.000)
p4@checkpoint: (-5.225, -12.019, -11.207, 17.244)
raw GEN: n/a
raw SIM: SimTrack #691 key=1765959

gamma (22)
Particle 261
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.068, -123.097, -124.355, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #692 key=1773375

gamma (22)
Particle 262
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000223
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000223
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.751, -29.370, 97.458, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #693 key=1773376

gamma (22)
Particle 263
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000386
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000386
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (63.703, 105.569, 8.792, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #696 key=1815938

gamma (22)
Particle 264
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001039
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001039
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-46.931, -114.019, -102.058, 0.000)
p4@checkpoint: (-0.000, -0.001, -0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #697 key=1823594

gamma (22)
Particle 265
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.001, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001978
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001978
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-47.038, -113.975, -102.689, 0.000)
p4@checkpoint: (-0.001, -0.001, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #698 key=1823595

gamma (22)
Particle 266
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000484
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000484
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-51.762, -111.909, -98.814, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #699 key=1823597

gamma (22)
Particle 267
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000085
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000085
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-43.897, -115.221, -103.854, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #700 key=1823610

gamma (22)
Particle 268
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000531
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000531
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-49.660, -112.857, -104.740, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #701 key=1823611

gamma (22)
Particle 269
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000308
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000308
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (119.189, 31.573, -190.213, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #702 key=1823613

gamma (22)
Particle 270
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000279
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000279
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-61.995, -106.581, -80.131, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #703 key=1823615

gamma (22)
Particle 271
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000298
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000298
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-44.202, -115.105, -103.686, 0.000)
p4@checkpoint: (0.000, -0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #704 key=1823619

e- (11)
Particle 272
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.012, -0.007, -0.001, 0.014)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #705 key=1823807

e- (11)
Particle 273
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.003, -0.001, 0.001, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #706 key=1824456

e- (11)
Particle 274
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, -0.003, -0.001, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #707 key=1824669

gamma (22)
Particle 275
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.182, 0.131, -0.040, 0.228)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 14 simE=0.227181
direct recHits: 0 missing=14 recoE=0.000000
subgraph simHits: 14 simE=0.227181
subgraph recHits: 0 missing=14 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.067, 72.039, -19.649, 0.000)
p4@checkpoint: (0.182, 0.131, -0.040, 0.228)
raw GEN: n/a
raw SIM: SimTrack #708 key=1825520

gamma (22)
Particle 276
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (67.454, 48.498, -14.700, 84.370)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 207 simE=83.401802
direct recHits: 0 missing=207 recoE=0.000000
subgraph simHits: 215 simE=83.406006
subgraph recHits: 0 missing=215 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.098, 71.995, -19.648, 0.000)
p4@checkpoint: (67.454, 48.498, -14.700, 84.370)
raw GEN: n/a
raw SIM: SimTrack #709 key=1825521

Bremsstrahlung
Vertex 207
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (97.936, 61.564, -18.935, 5.164)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #365 key=293

Annihilation

Vertex 208
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (97.506, 61.891, -18.493, 5.211)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #366 key=294

Bremsstrahlung
Vertex 209
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (98.109, 61.315, -20.042, 4.885)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #367 key=295

gamma (22)
Particle 277
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.606, 0.420, -0.131, 0.749)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 35 simE=0.719068
subgraph recHits: 0 missing=35 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #710 key=1825528

PairConversion

Vertex 205
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (96.343, 68.214, -18.731, 3.747)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #362 key=290

gamma (22)
Particle 278
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.071, 0.049, -0.015, 0.087)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 12 simE=0.057539
subgraph recHits: 0 missing=12 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #711 key=1825529

PairConversion

Vertex 200
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.560, 66.979, -18.347, 3.673)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #357 key=285

Bremsstrahlung
Vertex 201
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (95.745, 67.367, -18.907, 3.720)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #358 key=286

Bremsstrahlung
Vertex 202
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.099, 67.435, -18.036, 3.706)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #359 key=287

gamma (22)
Particle 279
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.303, 0.210, -0.065, 0.375)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 25 simE=0.370588
direct recHits: 0 missing=25 recoE=0.000000
subgraph simHits: 25 simE=0.370588
subgraph recHits: 0 missing=25 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.662, 71.204, -19.661, 0.000)
p4@checkpoint: (0.303, 0.210, -0.065, 0.375)
raw GEN: n/a
raw SIM: SimTrack #712 key=1825530

gamma (22)
Particle 280
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.032, 0.023, -0.007, 0.040)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.037142
direct recHits: 0 missing=6 recoE=0.000000
subgraph simHits: 6 simE=0.037142
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.662, 71.204, -19.661, 0.000)
p4@checkpoint: (0.032, 0.023, -0.007, 0.040)
raw GEN: n/a
raw SIM: SimTrack #713 key=1825531

e- (11)
Particle 281
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.051, 0.035, -0.011, 0.063)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 9 simE=0.054679
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #714 key=1865376

Bremsstrahlung
Vertex 203
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (94.977, 67.314, -18.442, 3.691)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #360 key=288

Bremsstrahlung
Vertex 204
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (95.216, 67.619, -18.513, 3.705)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #361 key=289

gamma (22)
Particle 282
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.000, -0.001, 0.002)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.001844
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.001844
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.605, 66.849, -22.919, 0.000)
p4@checkpoint: (0.002, -0.000, -0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #715 key=1865379

gamma (22)
Particle 283
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.001016
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.001016
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-122.928, 9.569, 124.198, 0.000)
p4@checkpoint: (-0.001, -0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #716 key=1865389

gamma (22)
Particle 284
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.037, 0.035, -0.009, 0.052)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.051926
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=0.051926
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.038, 72.079, -19.645, 0.000)
p4@checkpoint: (0.037, 0.035, -0.009, 0.052)
raw GEN: n/a
raw SIM: SimTrack #717 key=1865391

gamma (22)
Particle 285
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.002, -0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.002752
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.002752
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (99.850, 72.339, -21.610, 0.000)
p4@checkpoint: (0.002, 0.002, -0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #718 key=1865392

e- (11)
Particle 286
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.363, 0.251, -0.078, 0.448)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.431625
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 18 simE=0.431625
subgraph recHits: 0 missing=18 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.434, 71.525, -19.647, 0.000)
p4@checkpoint: (0.321, 0.300, -0.076, 0.446)
raw GEN: n/a
raw SIM: SimTrack #719 key=1865434

e+ (-11)
Particle 287
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.244, 0.168, -0.054, 0.301)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 19 simE=0.287077
direct recHits: 0 missing=19 recoE=0.000000
subgraph simHits: 20 simE=0.287443
subgraph recHits: 0 missing=20 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (100.948, 70.799, -19.700, 0.000)
p4@checkpoint: (0.266, 0.116, -0.054, 0.295)
raw GEN: n/a
raw SIM: SimTrack #720 key=1865435

Bremsstrahlung
Vertex 206
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (96.504, 68.848, -19.427, 3.800)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #364 key=292

gamma (22)
Particle 288
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000366
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000366
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-15.322, -122.344, -185.713, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #722 key=1865618

gamma (22)
Particle 289
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.002, 0.000, 0.001, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.002882
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.002882
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-98.706, 73.892, 96.557, 0.000)
p4@checkpoint: (-0.002, 0.000, 0.001, 0.003)
raw GEN: n/a
raw SIM: SimTrack #723 key=1877075

gamma (22)
Particle 290
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000226
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000226
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-102.605, -68.375, 188.661, 0.000)
p4@checkpoint: (-0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #724 key=1877080

gamma (22)
Particle 291
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000511
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000511
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (103.946, 66.318, -26.466, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #725 key=1877081

gamma (22)
Particle 292
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.001, -0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000576
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000576
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-68.359, -102.615, -214.680, 0.000)
p4@checkpoint: (-0.000, -0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #726 key=1877090

gamma (22)
Particle 293
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, -0.005, 0.004, 0.007)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.007873
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.007873
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-75.335, -97.609, 144.700, 0.000)
p4@checkpoint: (-0.004, -0.005, 0.004, 0.007)
raw GEN: n/a
raw SIM: SimTrack #727 key=2015388

n (2112)
Particle 294
pid: n (2112)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.145, 0.018, -0.196, 0.971)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.007331
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=0.007331
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-79.142, -94.549, 124.360, 0.000)
p4@checkpoint: (-0.145, 0.018, -0.196, 0.971)
raw GEN: n/a
raw SIM: SimTrack #728 key=2015395

gamma (22)
Particle 295
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.003, 0.000, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.003302
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.003302
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-63.097, -105.933, 136.871, 0.000)
p4@checkpoint: (0.002, -0.003, 0.000, 0.004)
raw GEN: n/a
raw SIM: SimTrack #729 key=2015396

connector

Particle 296
genEvent: 0
hasGen: no hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 1
nParents: 0 nChildren: 10
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 12326 simE=361.044586
subgraph recHits: 0 missing=12326 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 312 dE=789.252197
raw GEN: n/a
raw SIM: n/a

initial state

Vertex 213
initial state (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: no isSink: no
x4: (-0.002, 0.000, 2.166, -0.252)
nIn: 1 nOut: 10
raw GEN: n/a
raw SIM: n/a

interaction

Vertex 212
interaction (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: yes isSink: no
x4: (-0.002, 0.000, 2.166, -0.252)
nIn: 0 nOut: 1
raw GEN: n/a
raw SIM: n/a